



# Workshop: Leveraging the Adaptive Self-Configuration Features of ST's High-G IMU (separate registration required)

🕒 Thu, May 7 | 10:00 AM - 01:00 PM

📍 Great America 2



Part 1 hands-on with sensors

Part 2 demonstration with MCU

- High-G IMU Sensor Overview
- MEMS Studio Data Streaming and Visualization
- FSM Configuration, Test, and Visualization
- Implementing Adaptive Self-Configuration
- STM32C5 microcontroller introduction



# Leveraging the Adaptive Self-Configuration Features of ST's High-G IMU

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MEMS and Sensors Marketing and Application

Sensors Converge 2026

# Agenda hands-on with sensors

1 High G IMU Sensors

2 Hardware and Software Requirements

3 Hardware Overview

4 MEMS Studio GUI

5 Designing a basic FSM

6 Implementing Adaptive Self-Configuration (High-G)

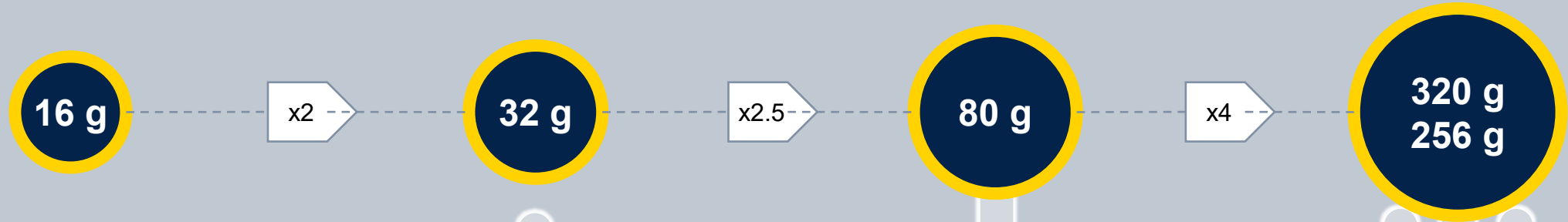
MLC: Machine Learning Core  
FSM: Finite State Machine  
IMU: Inertial Measurement Unit  
GUI: Graphical User Interface

# ST's High G IMU Sensors

IMU = Inertial Measurement Unit



# Continue to unleash the potential of our cutting-edge MEMS sensors



**Motion tracking,  
Walking, running, jogging**



**Medium intensity activity  
tracking**



**High-intensity activity  
tracking**



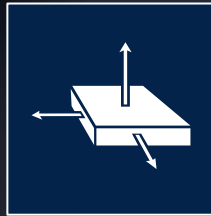
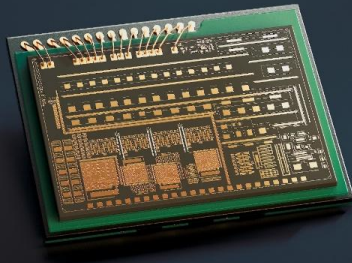
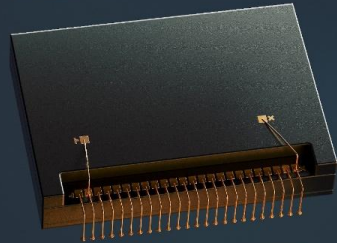
**High-g shock tracking**





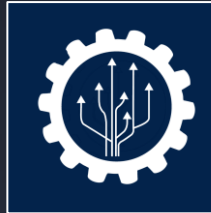
# The new industrial IMU: ISM6HG256X

**Intelligent, accurate IMU with simultaneous detection of low-g (16g) and high-g (256g) acceleration**



## **Top-notch accuracy & stability**

IMU with dual accelerometer full-scale designed to never miss any event, even the most severe



## **Processing at the edge and self-configurability**

Optimized performance and system level-power saving



## **Suitable for harsh environments**

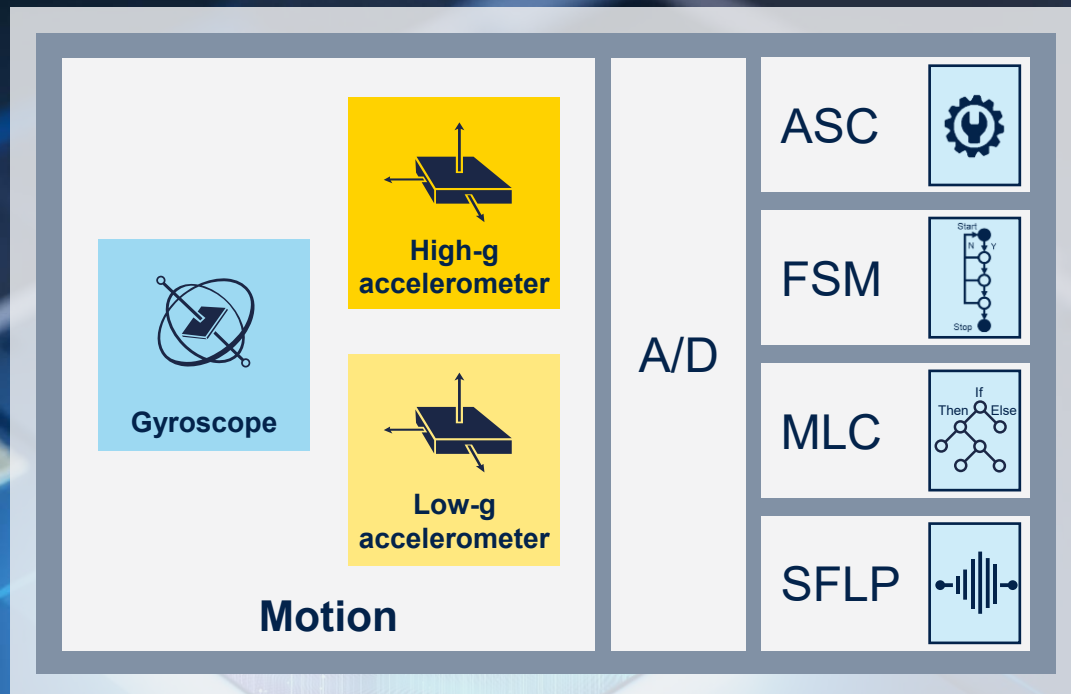
Operates reliably at temperatures up to 105°C





# What's inside the ISM6HG256X?

**A new sensor technology** that accurately, efficiently and simultaneously measures both high and low acceleration levels



**Consistent performance and valuable insights** for accurate sensing of both **subtle vibrations** and **severe shocks**

## Edge AI and self-configuration:

- Finite state machine (FSM)
- Machine learning core (MLC)
- Adaptive self-configuration (ASC)
- Sensor fusion low power (SFLP)



# ISM6HG256X

[link to product webpage](#)

## Intelligent, accurate IMU with simultaneous low-g and high-g acceleration detection

### 2 fully independent low-noise full-scale accelerometers:

Low-g accelerometer: up to 16g | 65  $\mu\text{g} / \sqrt{\text{Hz}}$

High-g accelerometer: up to 256g | 1 mg /  $\sqrt{\text{Hz}}$

**Low noise full-scale gyroscope**  
up to 4000 dps

### Intelligent sensor

MLC & FSM 2.0, ASC, SFLP 2.0,  
free fall & shock detection

### Low power consumption:

0.22 mA – Accelerometer low-g or high-g HP

0.8 mA – Combo (gyroscope + low-g & high-g accelerometer) HP

### Embedded smart FIFO

4.5 Kbyte

### Multiple interfaces

SPI, I2C, I3C 1.2 V compatible

### Extended operating temperature

-40 to +105 °C

### Compact package

LGA 2.5 x 3 x 0.71 14L



# A comprehensive software offering for your application

## Use cases

Shocks

Orientation

Tilt detection

Moving

Still detection

Free fall

Drop detection

Transport system detection

Position tracking



## Available software

- Smart asset tracking
- Car care (Towing, jacking, bump/crash)
- Activity recognition
- Fall / Hard-fall detection



Implemented  
on MLC / FSM\*

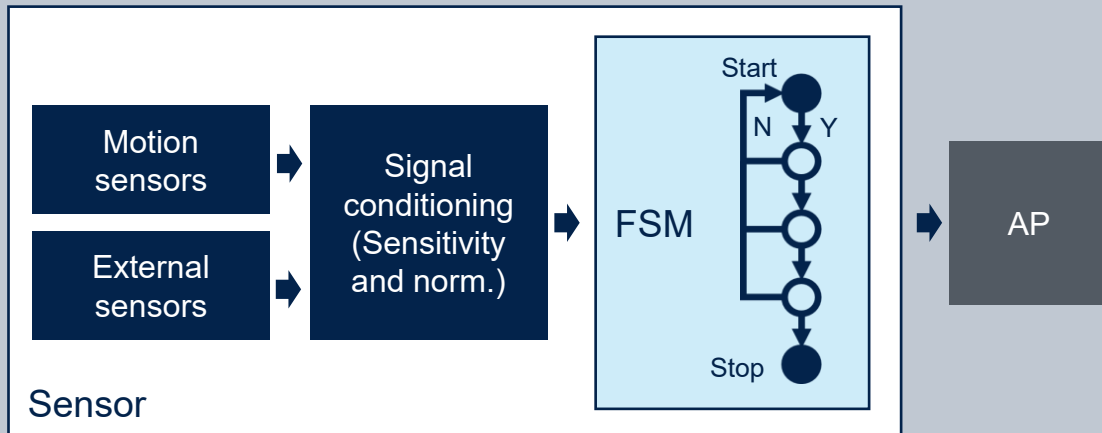
- MotionAC (accelerometer calibration)
- MotionAD (airplane detection)
- MotionFT (sliding DFT)
- MotionTL (tilt angles)
- MotionXLF (high-g, low-g fusion)
- MotionAT (active time)
- MotionFD (fall detection, fall height estimation)
- MotionGC (gyroscope calibration)



Software  
libraries

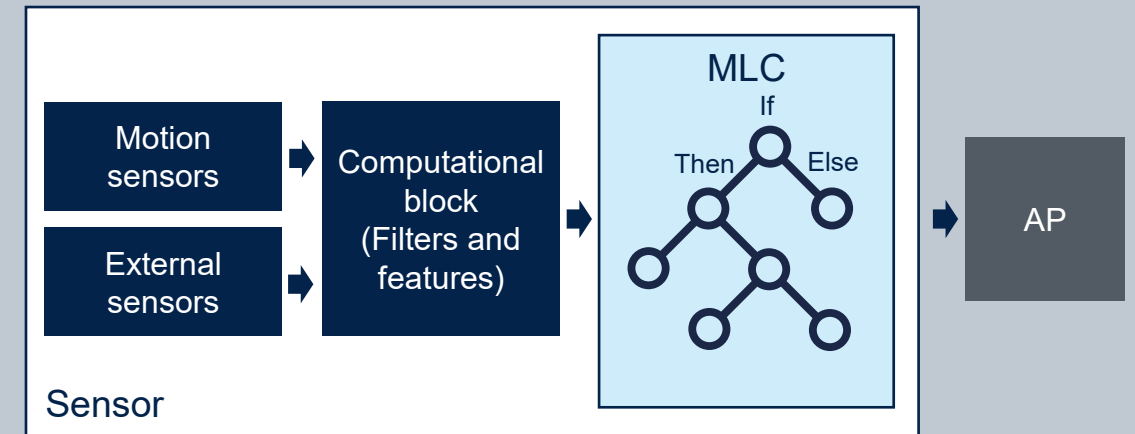
# Edge processing with MLC and FSM

## Finite state machine (FSM)



- FSM is composed of a finite number of user-defined states and transitions between them
- FSM can be in just one of the states and move to another one only if the transition condition is met
- Each instruction can be a command or a reset/next condition
- Used to detect sequential events using thresholds, timeouts, and MLC Core Results

## Machine learning core (MLC)



- The MLC runs classification models based on a decision tree logic: a binary tree of configurable nodes characterized by an “if-then-else” condition
- Decision tree is “built” offline through analysis of data sets
- It extract features from sensor data and feeds them to the decision trees to identify the current event/class

# Hardware and Software Requirements

# Hardware Requirements



Computer Running  
Windows 11



Mini USB Cable

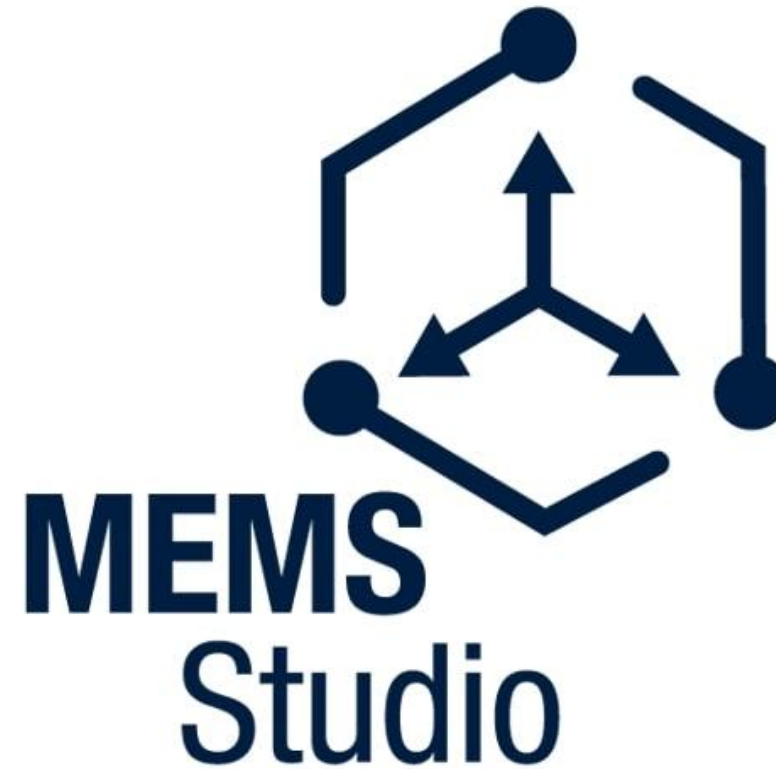


Nucleo F401



X-NUCLEO-IKS5A1

# Software Requirements

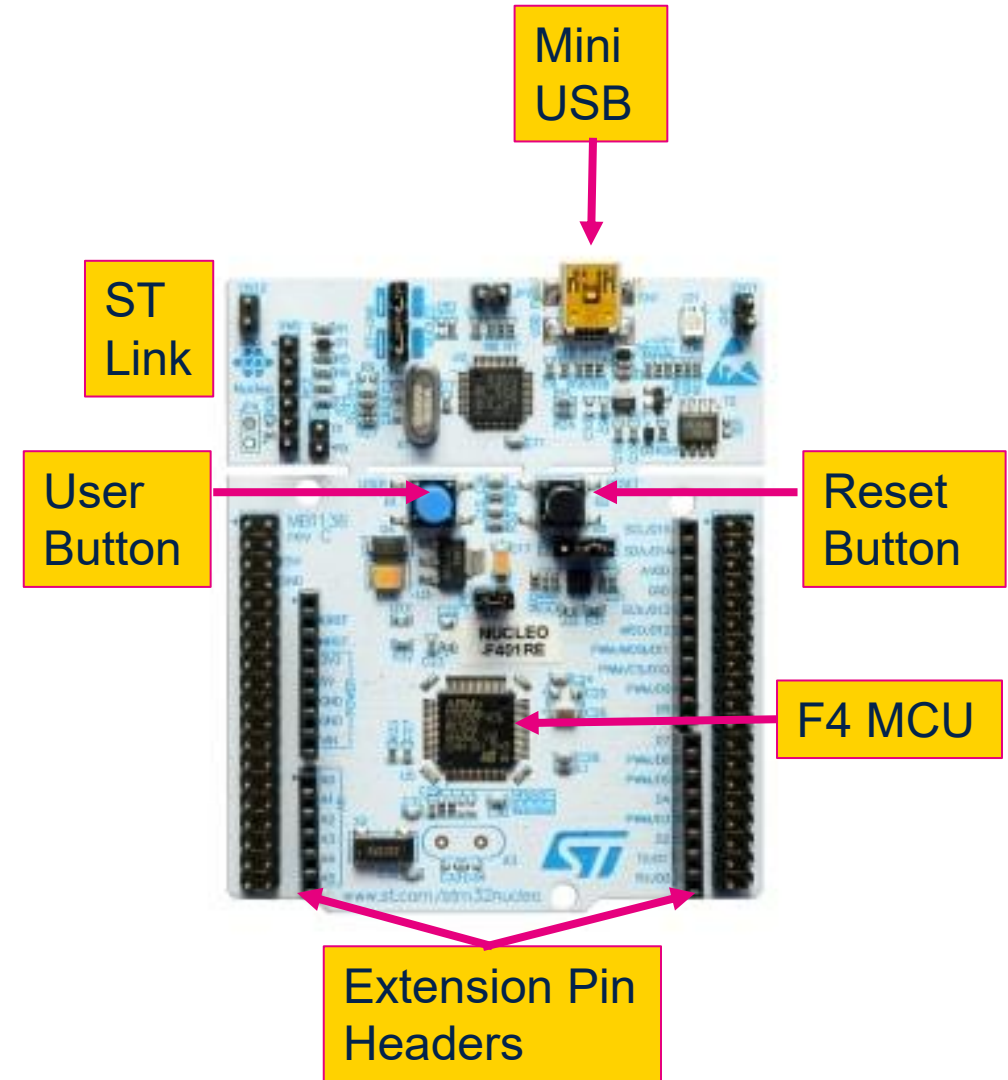




# Hardware Overview

# Nucleo-F401RE

- **STM32F401RE** microcontroller Cortex-M4 in LQFP64 package
- 1 user LED, 1 user push-button and 1 reset push-button
- Board expansion connectors
- On-board **ST-LINK/V2-1** debugger/programmer
- Three different interfaces supported on USB: mass storage, Virtual COM port and debug port
- Comprehensive free software libraries and examples available with the STM32Cube MCU Package



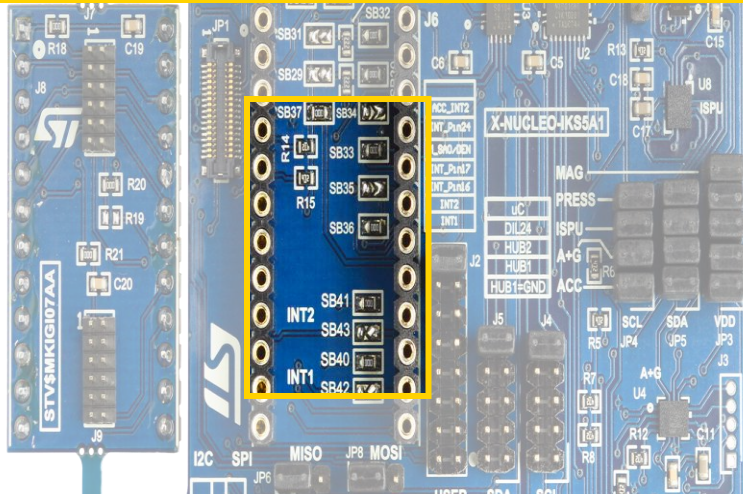


**Main board**  
X-NUCLEO-IND5A1

## Intelligent motion and environmental sensing platform

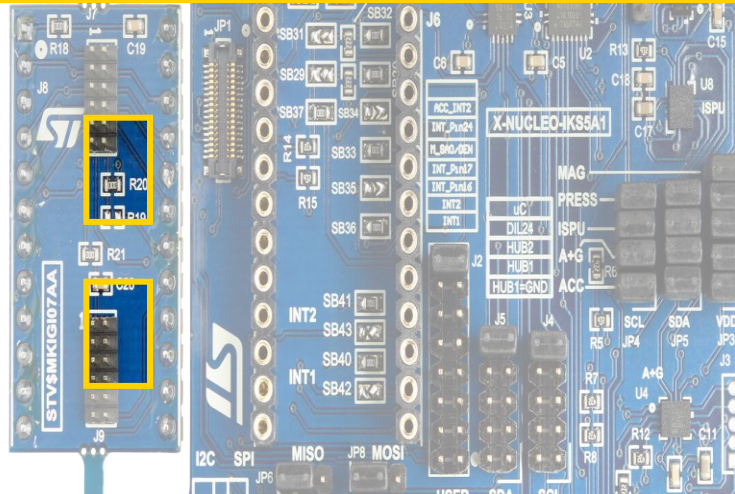
# External sensor connections for X-NUCLEO-IKS5A1

## DIL24 adapter



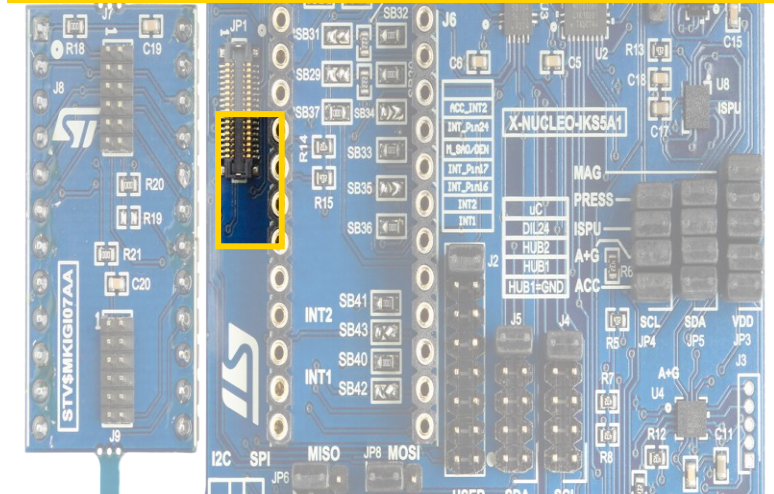
- One connector is available in the main board of X-NUCLEO-IKS5A1
- Standard DIL24 adapter for connecting add-on sensor boards directly to the main board

## Ribbon flat cable



- 2 connectors available on the detachable add-on board STEVAL-MKIGI07A
- Allows placing the sensor in a different position compared to being plugged into the main board
- Suitable for many industrial applications

## Flex PCB connection



- One connector is available in the main board of X-NUCLEO-IKS5A1
- Allows placing the sensor in a different position compared to being plugged into the main board
- Suitable for many industrial applications



# What is inside X-NUCLEO-IKS5A1

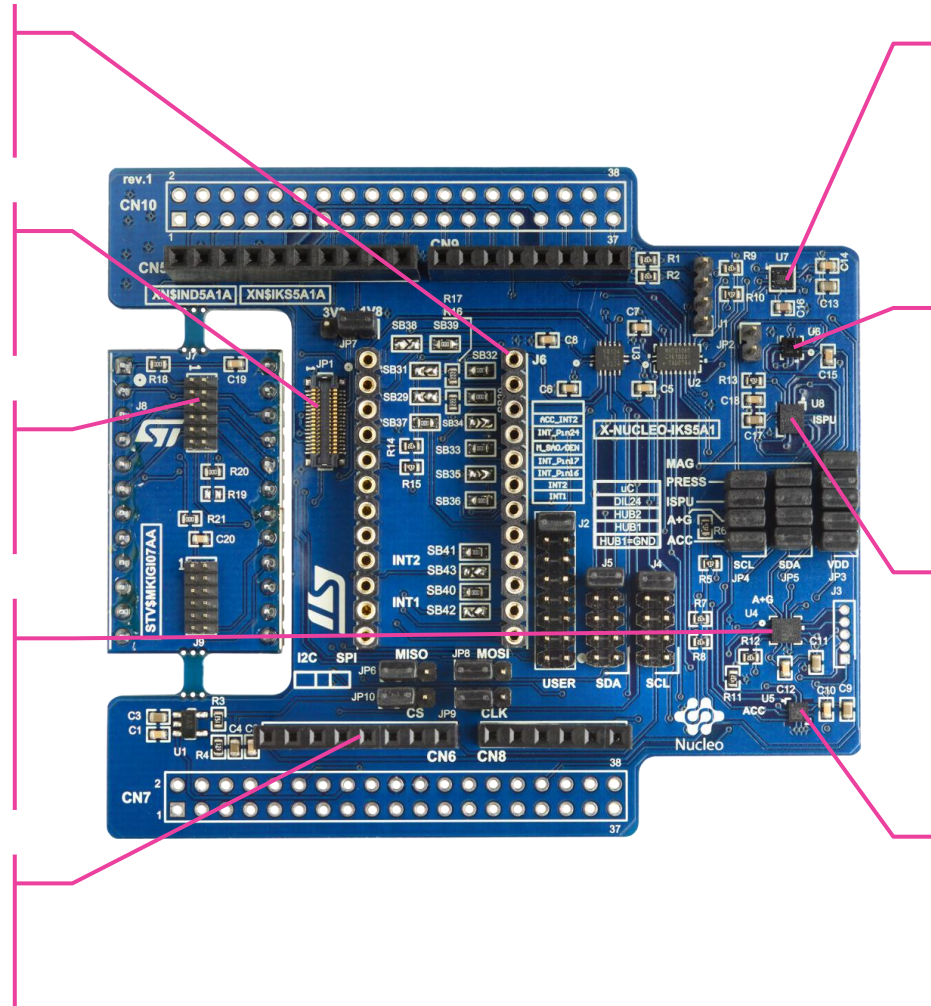
DIL24 socket for **external sensor**

Flex PCB connector for **external sensor**

Ribbon flat cable for **external sensor**

**Intelligent IMU** with simultaneous  
**low-g & high-g acceleration**  
detection - **ISM6GH256X**

Arduino® UNO R3 connector



**High accuracy, ultralow power,**  
**3-axis digital magnetometer -**  
**IIS2MDC**

**Dual full-scale, 1260 hPa and**  
**4060 hPa, absolute digital output**  
**barometer - ILPS22QS**

**6-axis IMU**, always-on 3-axis  
accelerometer and 3-axis  
gyroscope **with ISPU - ISM330IS**

**Intelligent ultralow power**  
**accelerometer** for industrial  
applications - **IIS2DULPX**



# **MEMS Studio**



HANDS  
ON

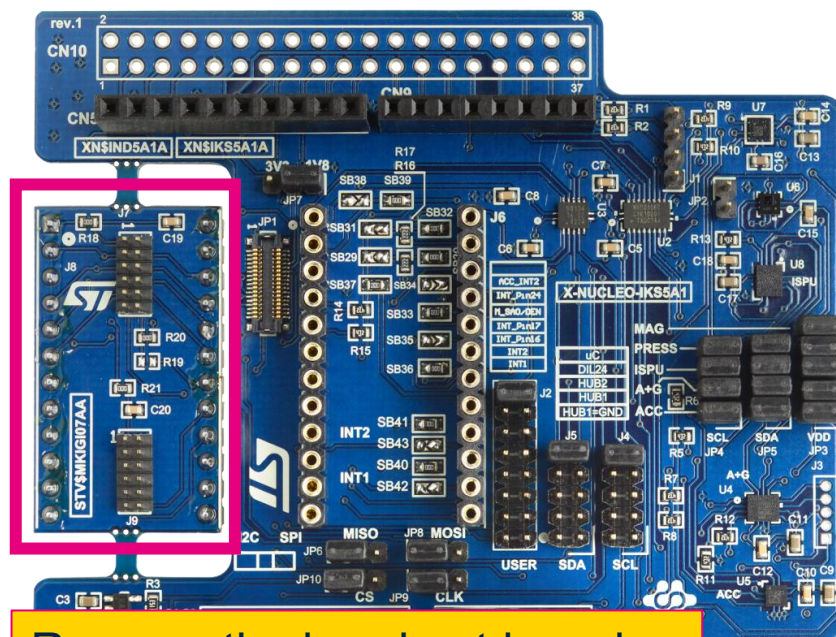
# Hands On

- When ever you see the HANDS ON sticker you should follow along using your own hardware and software
- There will also be written instructions in case you get lost or fall behind

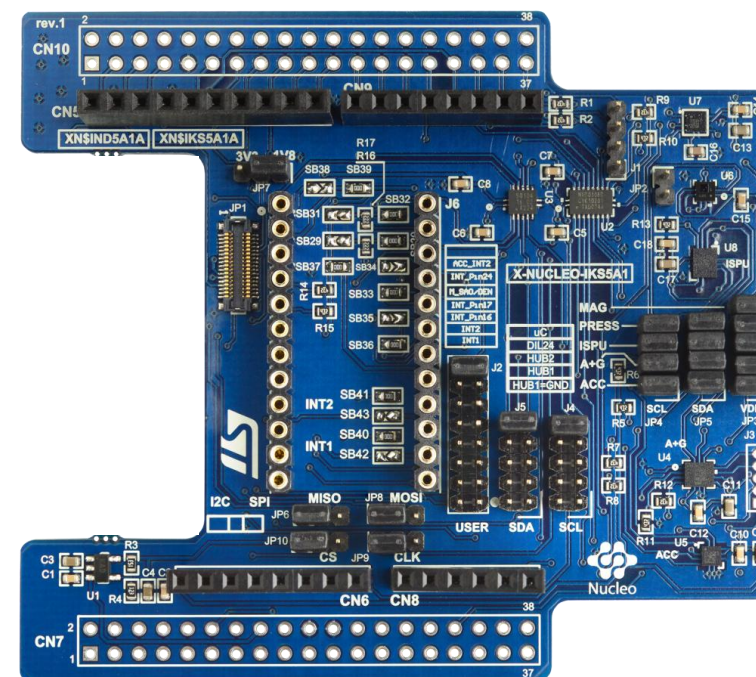
Written Instructions  
show up in these boxes

HANDS  
ON

# Hardware Setup



Remove the breakout board so that it doesn't accidentally press down on the user input or reset buttons on the Nucleo



# HANDS ON

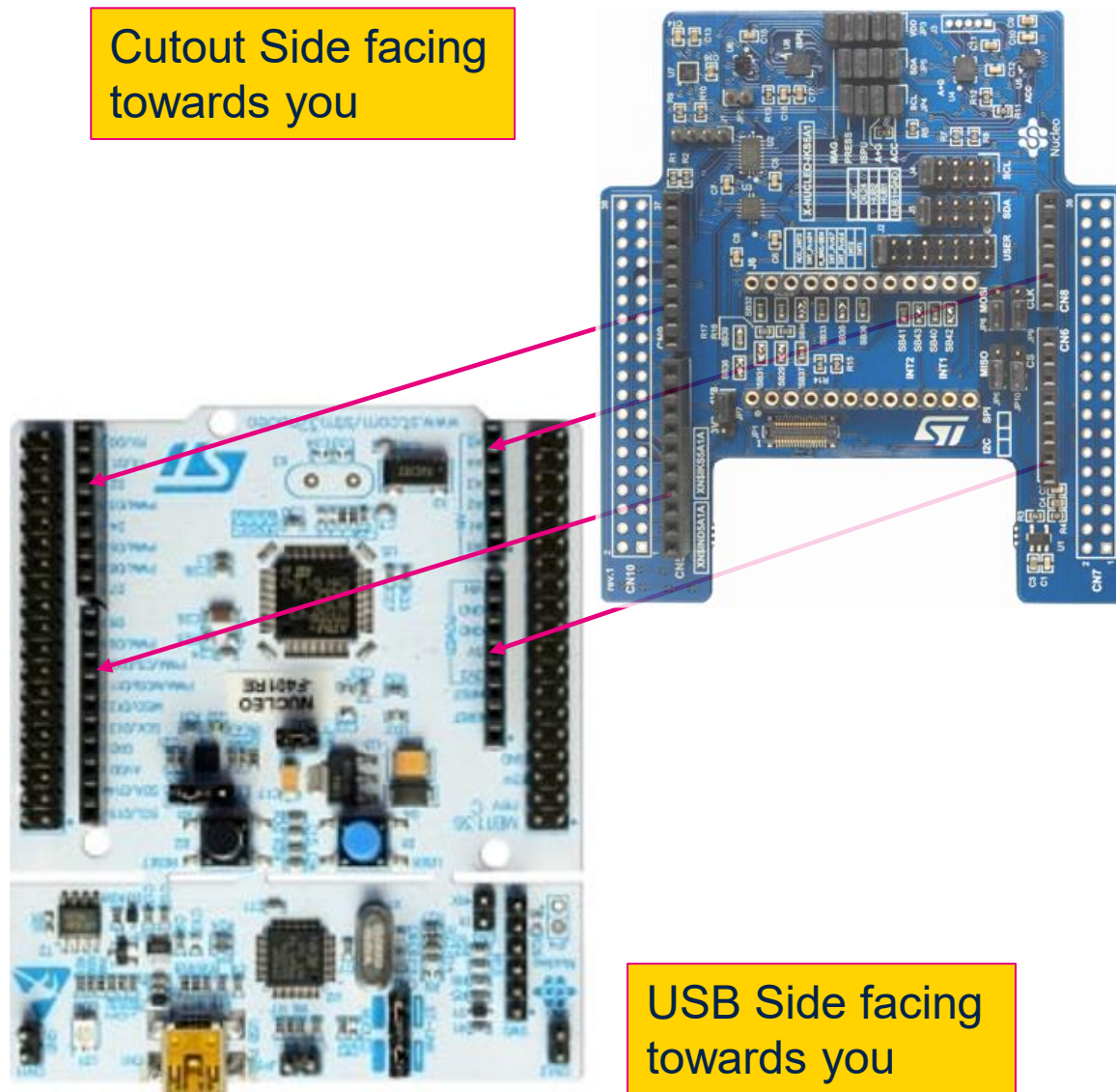
## Hardware Setup

Connect the X-Nucleo-IKS5A1 to the top of the Nucleo-F401RE using the pins on the bottom of the X-Nucleo board

Orientation shown to the right

Once all pins are lined up press down to confirm the connection. Should fit snugly with only a small gap

Cutout Side facing  
towards you

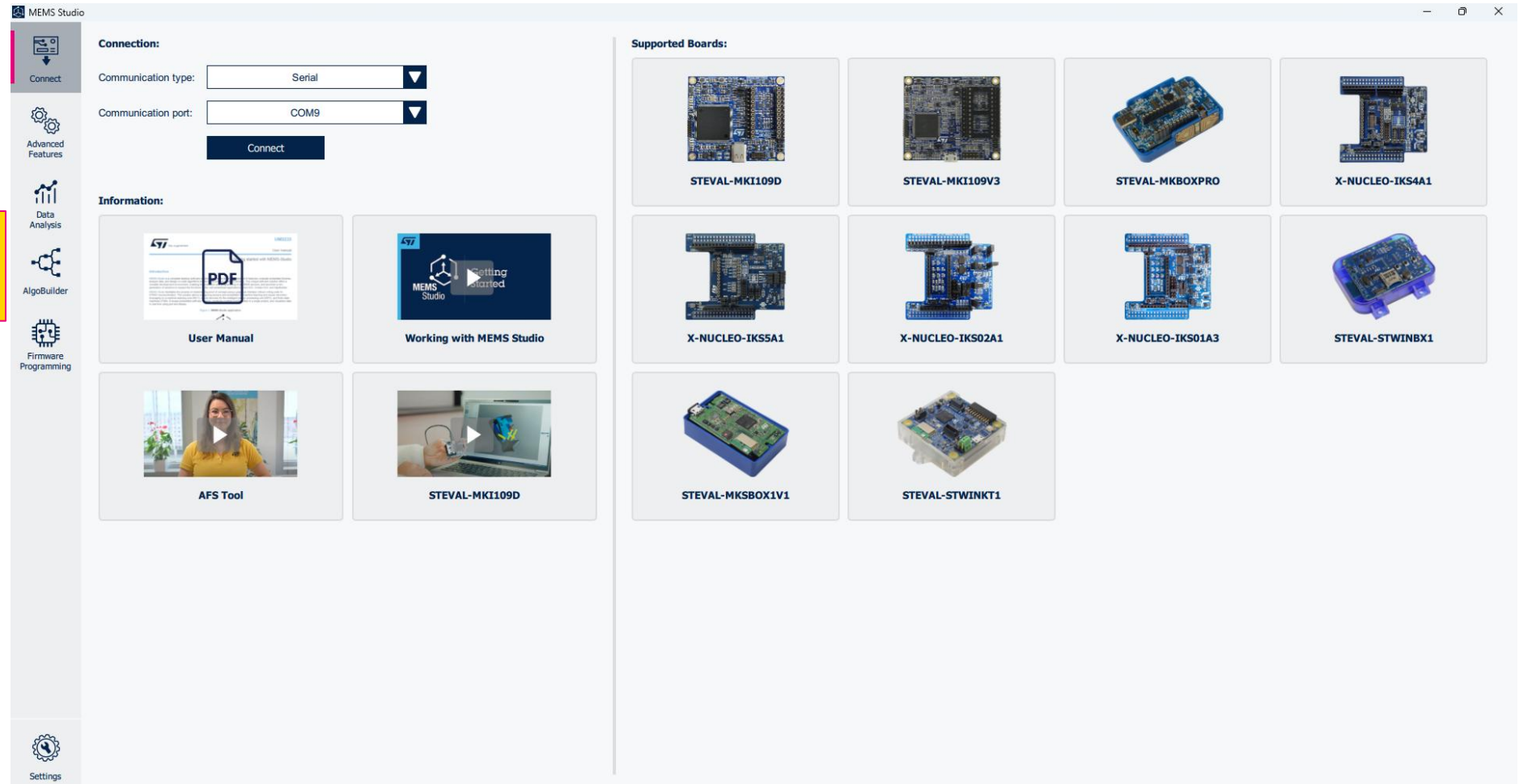


USB Side facing  
towards you

HANDS  
ON

Launch MEMS Studio  
GUI

# Application Setup

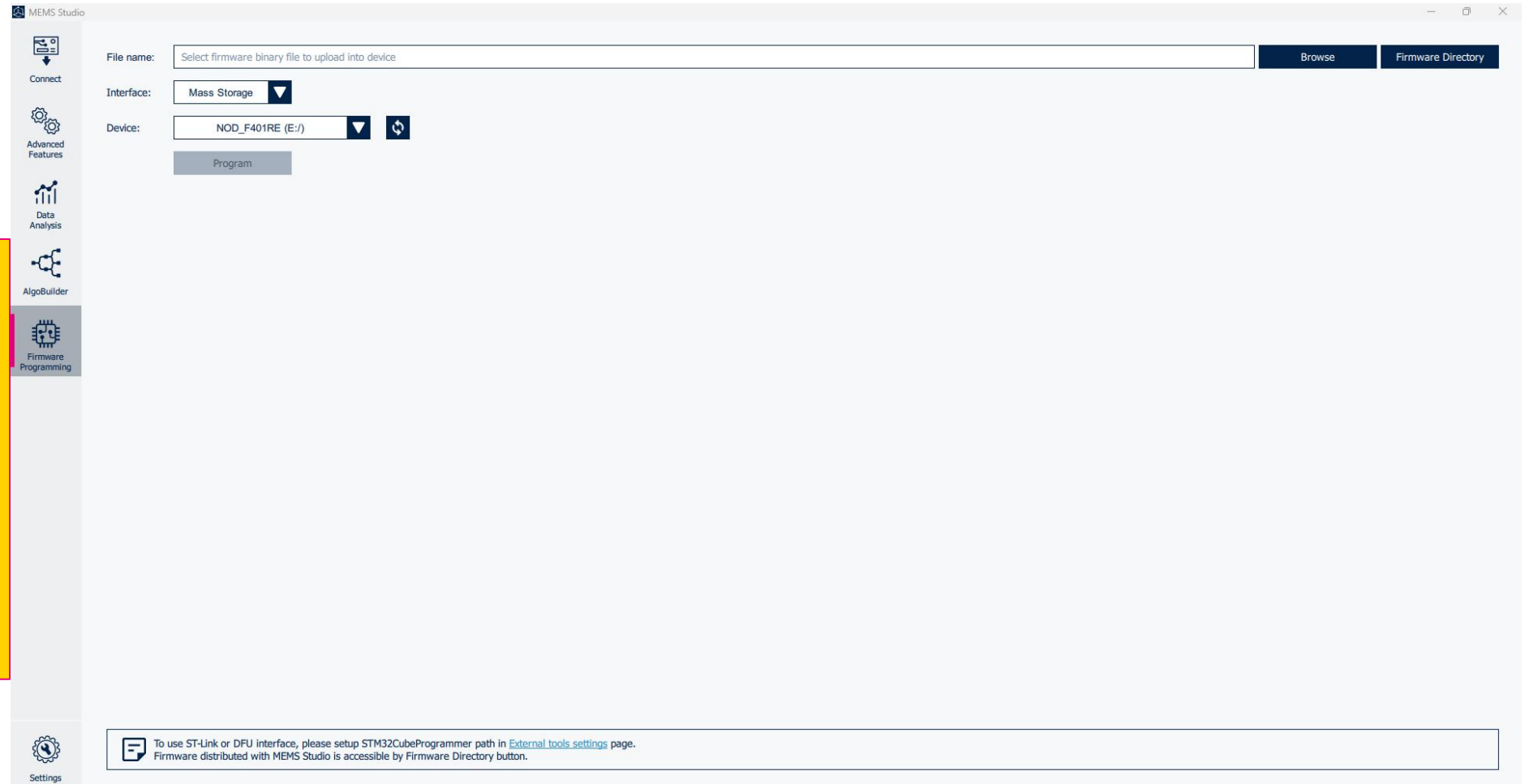




HANDS  
ON

Navigate to the  
Firmware Programming  
tab found along the left-  
hand side of the UI

Then click on the  
Firmware Directory  
Button found in the top  
right of the UI



# HANDS ON

## Program the Device

Navigate to the corresponding firmware for the workshop based on the hardware.

X-Nucleo-IKS5A1->  
NucleoF401Re ->  
DataLogExtended.bin  
and select and open.



# HANDS ON

Make sure the interface is selected as Mass Storage and your device appears.

Then click Program to begin flashing the board.

## Program the Device

|            |                  |    |
|------------|------------------|----|
| Interface: | Mass Storage     | ▼  |
| Device:    | NOD_F401RE (E:/) | ▼  |
|            |                  | ↺↻ |
| Program    |                  |    |

# Board Connection

HANDS  
ON

**Connection:**

Communication type:

Communication port:

**Sensor(s):**

|                       |                                                                            |                     |                                                                            |
|-----------------------|----------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------|
| Accelerometer sensor: | <input type="text" value="ISM6HG256X"/> <input type="button" value="New"/> | Gyroscope sensor:   | <input type="text" value="ISM6HG256X"/> <input type="button" value="New"/> |
| Magnetometer sensor:  | <input type="text" value="IIS2MDC"/> <input type="button" value="Active"/> | Pressure sensor:    | <input type="text" value="ILPS22QS"/> <input type="button" value="New"/>   |
| Humidity sensor:      | <input type="text" value="None"/>                                          | Temperature sensor: | <input type="text" value="None"/>                                          |

HANDS  
ON

# Sensor Configuration

The screenshot displays the ST Sensor Configuration tool interface. On the left is a sidebar with navigation options: Connect, Sensor Evaluation (highlighted), Advanced Features, Data Analysis, AlgoBuilder, Firmware Programming, and Settings. The main area is titled 'Sensor Evaluation' and contains a 'Reset Timestamp' button at the top. Below this are configuration sections for several sensors:

- ISM6HG256X Low-g (ACCELEROMETER)**
  - Accelerometer low-g Full Scale: 8 g
  - Accelerometer low-g Output Data Rate: 960 Hz (HP)
- ISM6HG256X High-g (ACCELEROMETER)**
  - Accelerometer high-g Full Scale: 256 g
  - Accelerometer high-g Output Data Rate: 960 Hz
- ISM6HG256X (GYROSCOPE)**
  - Gyroscope Full Scale: 2000 dps
  - Gyroscope Output Data Rate: 960 Hz (HP)
- IIS2MDC (MAGNETOMETER)**
  - Magnetometer Operating Mode: High resolution
  - Magnetometer Output Data Rate: 100 Hz
  - Magnetometer System Operating Mode: Continuous
  - Magnetometer LP Filter: Disabled [BW=ODR/2]
  - Offset cancellation: ☐
- ILPS22QS (PRESSURE SENSOR)**
  - Pressure Output Data Rate: 100 Hz
  - Buttons: Set AUTOZERO, Reset AUTOZERO
- Data stream configuration**
  - Interrupts: ☐
  - Finite State Machine: ☐
  - Machine Learning Core: ☐



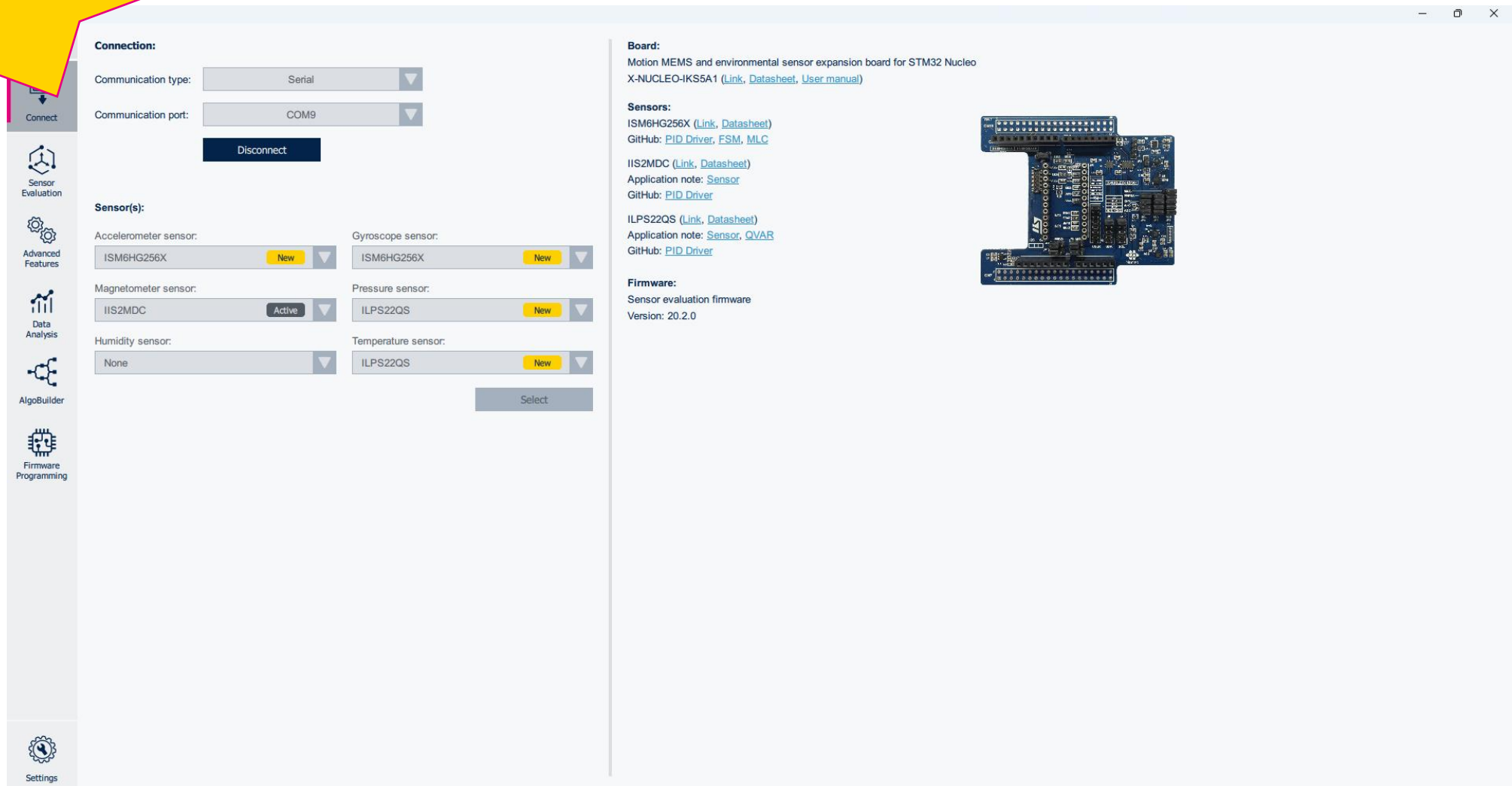
HANDS  
ON

# Data Streaming



HANDS  
ON

# Disconnect Board



**Connection:**

Communication type:

Communication port:

**Disconnect**

**Sensor(s):**

Accelerometer sensor:  **New**

Gyroscope sensor:  **New**

Magnetometer sensor:  **Active**

Pressure sensor:  **New**

Humidity sensor:

Temperature sensor:  **New**

**Select**

**Board:**

Motion MEMS and environmental sensor expansion board for STM32 Nucleo X-NUCLEO-IKS5A1 ([Link](#), [Datasheet](#), [User manual](#))

**Sensors:**

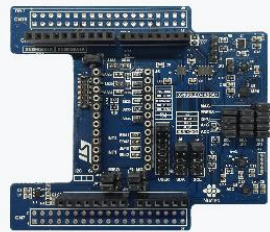
ISM6HG256X ([Link](#), [Datasheet](#))  
GitHub: [PID Driver](#), [FSM](#), [MLC](#)

IIS2MDC ([Link](#), [Datasheet](#))  
Application note: [Sensor](#)  
GitHub: [PID Driver](#)

ILPS22QS ([Link](#), [Datasheet](#))  
Application note: [Sensor](#), [QVAR](#)  
GitHub: [PID Driver](#)

**Firmware:**

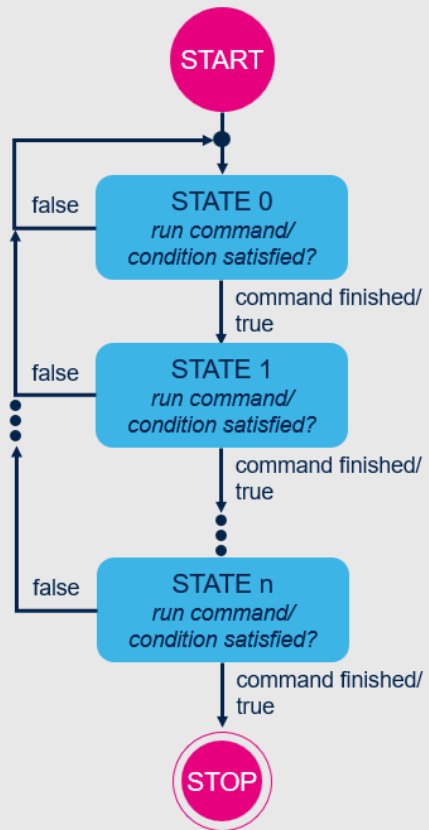
Sensor evaluation firmware  
Version: 20.2.0



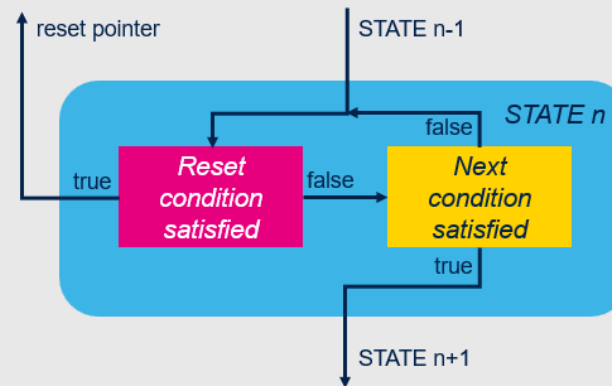
# Designing a basic FSM

# FSM in general

## Generic finite state machine



## Reset/Next state type



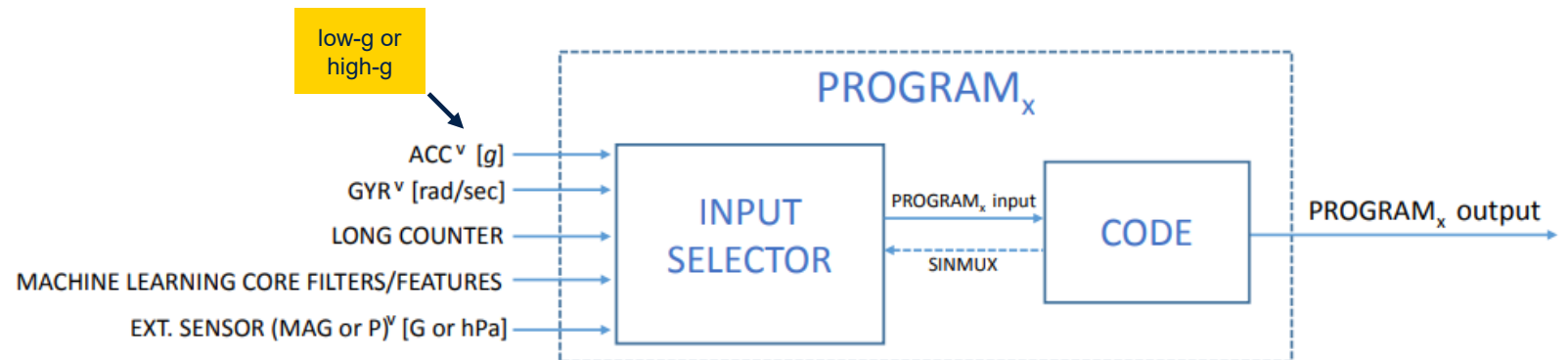
2 types of states:

- Command
  - perform special tasks for flow control, output, and synchronization
  - executed as one single-step command (no need for data sample)
- Condition (RNC – Reset/Next Condition)
  - a combination of two conditions, that are used to reset or continue the program flow

# Input of FSM

Selected by **SINMUX** command (optionally followed by EXT\_SINMUX)

- Default is low-g accelerometer data
- Available input sources
  - Low-g accelerometer, with precomputed norm (V)
  - High-g accelerometer, with precomputed norm (V)
  - Gyroscope, with precomputed norm (V)
  - External sensor (Pressure / Magnetometer, optionally other)
  - Filtered signal from the MLC
  - MLC feature
  - Integrated gyroscope signal
  - Long counter value



HANDS  
ON

Advanced Features

Data Analysis

AlgoBuilder

Firmware Programming

Settings

Connection:

Communication type: Serial

Communication port: COM9

Disconnect

Sensor(s):

Accelerometer sensor: ISM6HG256X New

Gyroscope sensor: ISM6HG256X New

Magnetometer sensor: IIS2MDC Active

Pressure sensor: ILPS22QS New

Humidity sensor: None

Temperature sensor: None

Select



HANDS  
ON

# Sensor Configuration

The screenshot displays the ST Sensor Configuration software interface. On the left is a sidebar with navigation options: Connect, Sensor Evaluation, Advanced Features, Data Analysis, AlgoBuilder, Firmware Programming, and Settings. The main area is titled 'Easy Configuration' and 'Reset Timestamp'. It contains five sensor configuration sections, each with a title bar and a list of settings with dropdown menus or checkboxes.

- ISM6HG256X Low-g (ACCELEROMETER)**
  - Accelerometer low-g Full Scale: 8 g
  - Accelerometer low-g Operating Mode: High-performance mode
  - Accelerometer low-g Output Data Rate: 240 Hz
- ISM6HG256X High-g (ACCELEROMETER)**
  - Accelerometer high-g Full Scale: 32 g
  - Accelerometer high-g Output Data Rate: Power-Down
- ISM6HG256X (GYROSCOPE)**
  - Gyroscope Full Scale: 2000 dps
  - Gyroscope Operating Mode: High-performance mode
  - Gyroscope Output Data Rate: 120 Hz
- IIS2MDC (MAGNETOMETER)**
  - Magnetometer Operating Mode: High resolution
  - Magnetometer Output Data Rate: 100 Hz
  - Magnetometer System Operating Mode: Continuous
  - Magnetometer LP Filter: Disabled [BW=ODR/2]
  - Offset cancellation: ☒
- ILPS22QS (PRESSURE SENSOR)**
  - Pressure Full Scale: 1260 hPa
  - Pressure Output Data Rate: 25 Hz
  - Pressure Averaging: Prs Avg = 4
  - Pressure LP filter configuration: Enabled [BW=ODR/4]
  - Buttons: Set AUTOZERO, Reset AUTOZERO

At the bottom, the status bar shows 'Version: 2.3.1' and 'Connected: X-NUCLEO-IKS5A1, FW: 20.2.0'.

# HANDS ON

# FSM Tool

To navigate to the FSM tool go to the

Advanced Features tab on the left side of the UI and then

Click on the FSM option

The screenshot shows the MEMS Studio interface with the FSM Tool selected. The left sidebar contains the 'Advanced Features' tab, which is expanded to show the 'FSM' option. The main window displays the 'FSM parameters' section, which includes a 'Sensor' dropdown (ISM6HG256X), a 'State machine' dropdown (#1), and an 'ODR' dropdown (30 Hz). Below these are 'Converter' options for 'Float' (1.00) and 'Float16' (3C00). The 'Long Counter' section has a 'Max value' of 0000 and 'Interrupts' for INT1 and INT2. The 'FSM start address' section has 'FSM\_START\_H' (04) and 'FSM\_START\_L' (00). The 'Status Data' section has checkboxes for 'Enabled', 'INT1', and 'INT2'. The 'Fixed Data' section has a checkbox for 'Decimation'. The bottom of the window has a 'Configuration' tab and a 'Testing' tab.

**MEMS Studio**

**Advanced Features**

- Pedometer
- FSM**
- MLC
- Merge MLC + FSM
- ISPU IDE
- ISPU Model Converter
- IR Algorithms Tuning
- Configuration Converter
- HSDatalog Converter

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 30 Hz

Converter: Float: 1.00 Float To Float16 Float16: 3C00 Float16 To Float

Long Counter: Max value: 0000 Interrupts: ☐ INT1 ☐ INT2

FSM start address: FSM\_START\_H: 04 FSM\_START\_L: 00

**Status Data**

- Enabled ☐
- INT1 ☐
- INT2 ☐

**Fixed Data**

- Decimation ☐

**Index Address Type**

**Configuration Testing**

HANDS  
ON

# Set ODR

First step is to set the  
FSM Cores ODR.

In this example we will  
be using **240Hz**

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 240 Hz

MEMS Studio

**Advanced Features**

- Pedometer
- FSM
- MLC
- Merge MLC + FSM
- ISPU IDE
- ISPU Model Converter
- IR Algorithms Tuning
- Configuration Converter
- HSDatalog Converter

Connect

Sensor Evaluation

Advanced Features

Data Analysis

AlgoBuilder

Firmware Programming

Settings

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 240 Hz

Converter

Float: 1.00 Float To Float16

Float16: 3C00 Float16 To Float

Long Counter

Max value: 0000

Interrupts: ☐ INT1 ☐ INT2

FSM start address

FSM\_START\_H: 04

FSM\_START\_L: 00

Status Data

Enabled ☐

INT1 ☐

INT2 ☐

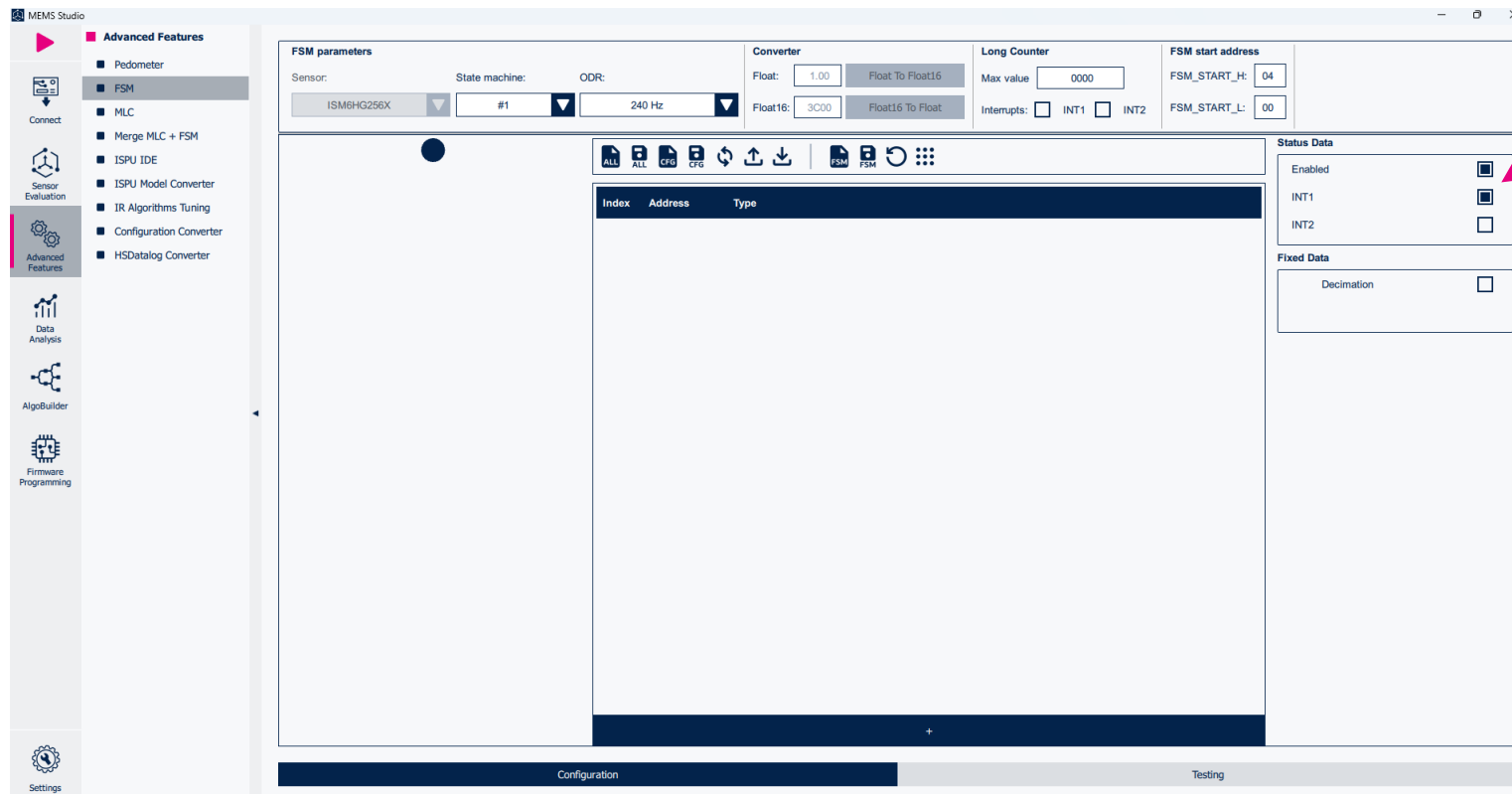
Fixed Data

Decimation ☐

Configuration Testing

HANDS  
ON

# Enable Interrupt and Status



**Status Data**

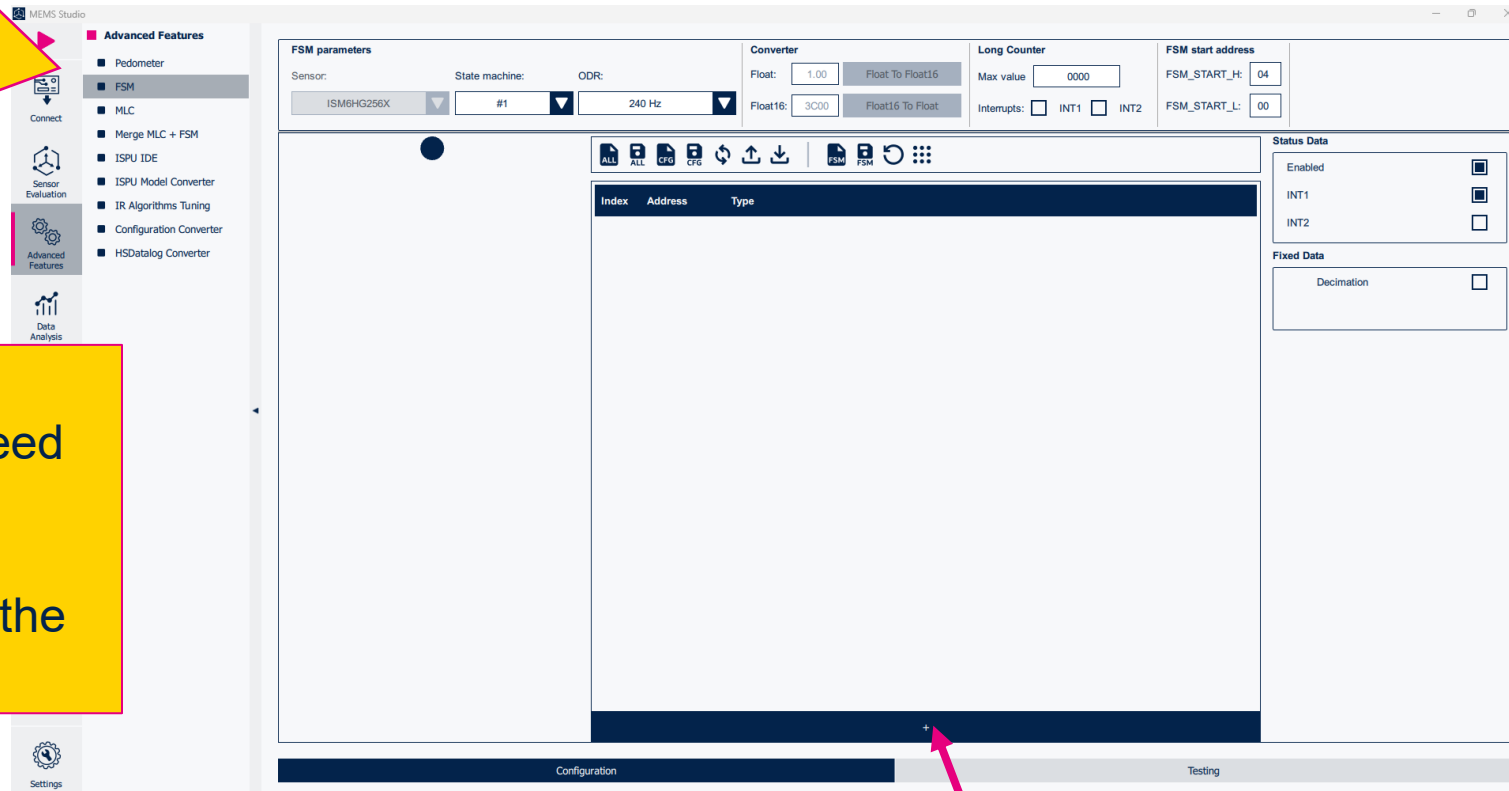
|         |                                     |
|---------|-------------------------------------|
| Enabled | <input checked="" type="checkbox"/> |
| INT1    | <input checked="" type="checkbox"/> |
| INT2    | <input type="checkbox"/>            |

Next, we will Enable the FSM Status and drive the interrupt to the INT1 Interrupt pin

# HANDS ON

## Add New State

To start creating our FSM Program we need to add our First Instruction. To do so click the + button at the bottom of the UI



HANDS  
ON

# Add Reset Next Condition for Motion

| Index | Address | Type  |       |         |                                                                                                                                                                                                                                                             |
|-------|---------|-------|-------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S0    | 0x0A    | RNC ▼ | NOP ▼ | GNTH1 ▼ |    |

Instructions can either be Commands or Reset/Next Conditions. In this case we will set the type to **RNC** for Reset/Next Condition.

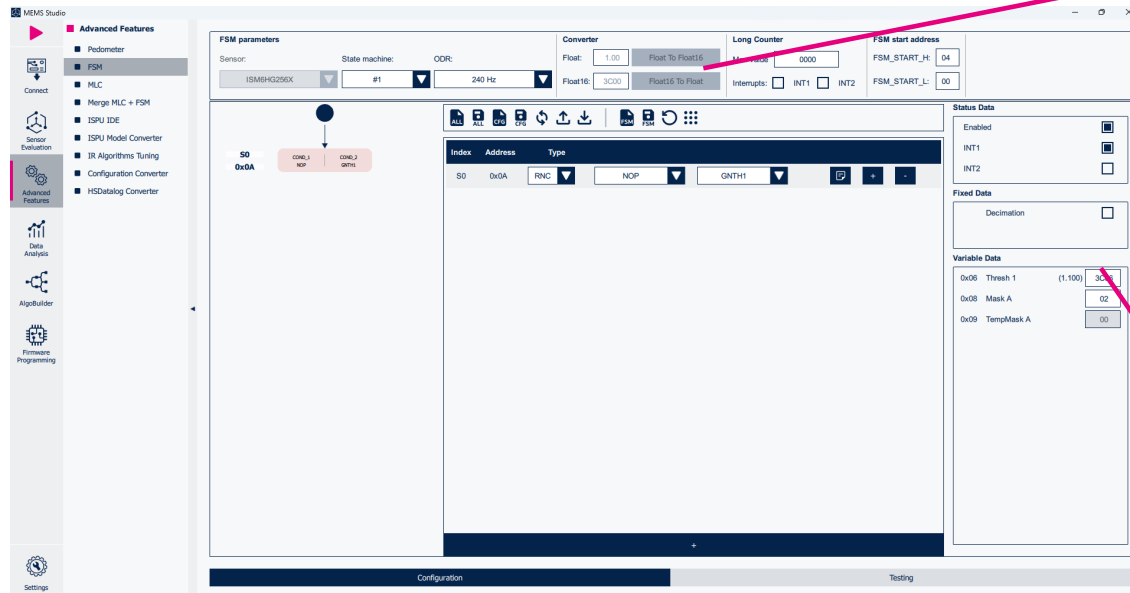
The RNC is split into two sections. The Reset and Next conditions. For our Reset condition we will use the No operation condition or **NOP**

This will loop forever until the Next condition is true. In this case we will use the Greater than Threshold 1 or **GNTH1**



HANDS  
ON

# Update Threshold Using Converter



**Converter**

Float:

Float16:

Now we need to set the value for the variable Thresh 1. We want to use 1.1g as the threshold. The Variable however is expecting a Float16 Hex value.

**Variable Data**

|      |          |         |                                   |
|------|----------|---------|-----------------------------------|
| 0x06 | Thresh 1 | (1.100) | <input type="text" value="3C66"/> |
|------|----------|---------|-----------------------------------|

You can generate this using the converter at the top of the UI and coping the result into the variable Section. You will see the UI update with the converted value.

# Set Mask A to V+

HANDS  
ON

Set Mask A

X+ | X- | Y+ | Y- | Z+ | Z- | V+ | V- = 0000 0010 = 02  
Binary Hex

Next, we want to set the Mask Value. We can utilize If you hover over the Mask A variable it will show the mask selection for the current part.

## Variable Data

|      |          |         |      |
|------|----------|---------|------|
| 0x06 | Thresh 1 | (1.100) | 3C66 |
| 0x08 | Mask A   |         | 02   |

We want to use the Norm of all the axis so this works regardless of orientation and motion. Therefore, we want to use **V+** or **02** in Hex

HANDS  
ON

# Add New State

MEMS Studio

**Advanced Features**

- Pedometer
- FSM
- MLC
- Merge MLC + FSM
- ISPU IDE
- ISPU Model Converter
- IR Algorithms Tuning
- Configuration Converter
- HSDatalog Converter

Connect

Sensor Evaluation

Advanced Features

Data Analysis

AlgoBuilder

Firmware Programming

Settings

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 240 Hz

Converter: Float: 1.00 Float To Float16 Float16: 3C00 Float16 To Float

Long Counter: Max value: 0000 Interrupts: ☐ INT1 ☐ INT2

FSM start address: FSM\_START\_H: 04 FSM\_START\_L: 00

State machine diagram: S0 (0x0A) → COND\_1 (NOP) → COND\_2 (GNT1)

| Index | Address | Type |
|-------|---------|------|
| S0    | 0x0A    | RNC  |
|       |         | NOP  |
|       |         | GNT1 |

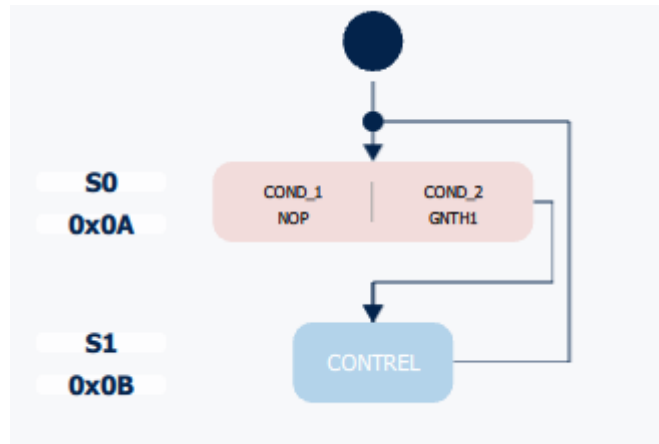
Configuration Testing

Let us add another  
Instruction



HANDS  
ON

# Set to Command CONTROL



| Index | Address | Type  |           |         |  |  |  |  |
|-------|---------|-------|-----------|---------|--|--|--|--|
| S0    | 0x0A    | RNC ▼ | NOP ▼     | GNTH1 ▼ |  |  |  |  |
| S1    | 0x0B    | CMD ▼ | CONTROL ▼ |         |  |  |  |  |

We will use the command  
CONTROL.

This resets the program from  
the Reset Pointer and  
generates an Interrupt

# About the Saves



## Save/Load All state machines

This saves all Current State machines to a .json file configuration

## Save/Load Device Configuration

This saves the current device configuration plus all active state machines to a .json file configuration

## Write/Load FSM Configuration to sensor

This write/reads current state machine information to the device.

## Save/Load State Machine

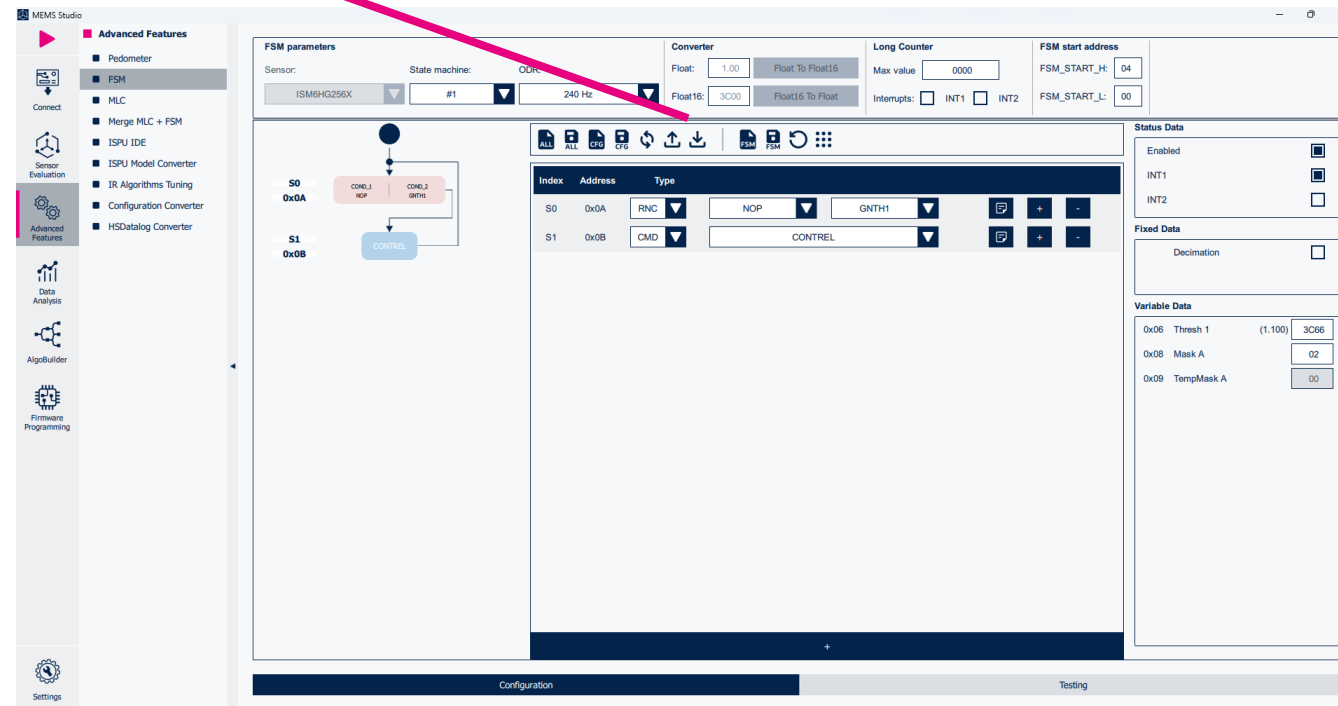
Saves current state machine only. Also, can load legacy .fsm files.

HANDS  
ON

# Save Configuration



We now can save the  
configuration

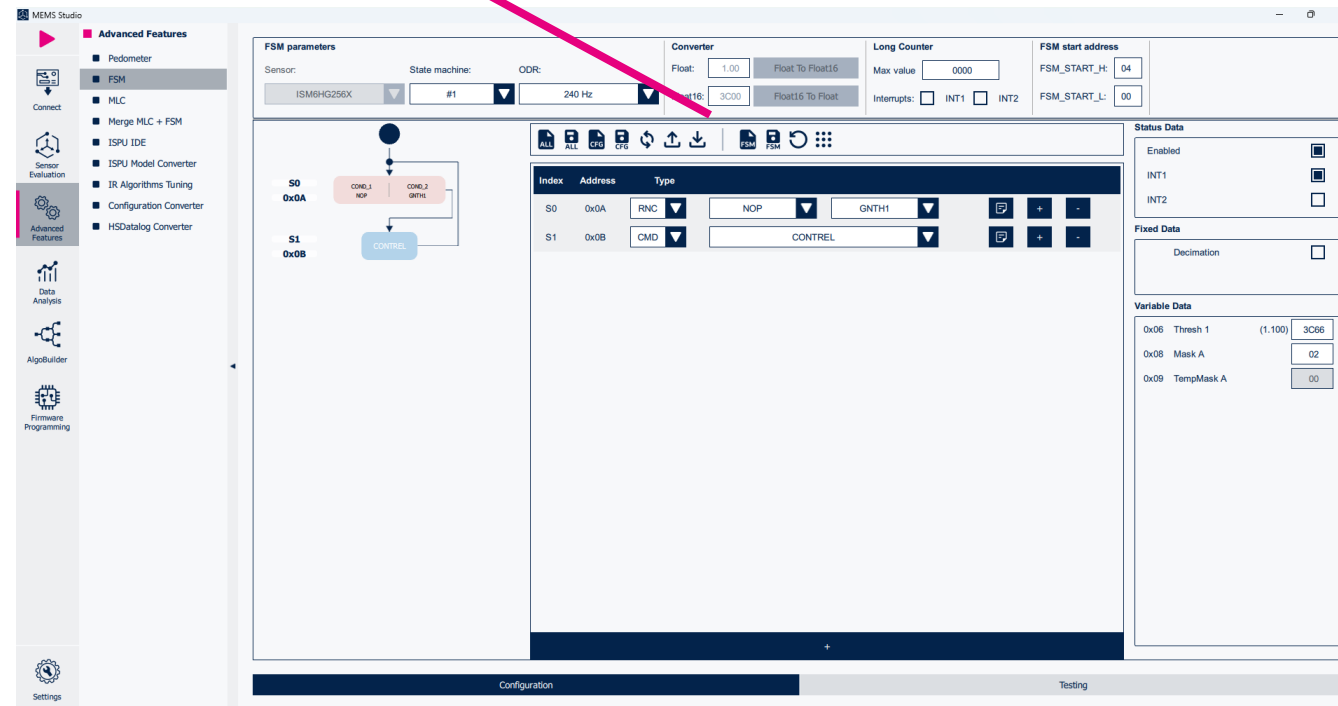


HANDS  
ON

# Write Configuration to Sensor



And write the  
configuration to the  
device





# Start Data Streaming

HANDS  
ON

Hit the play button in the top right of the UI to begin streaming data

The screenshot displays the MEMS Studio software interface. On the left, a sidebar contains icons for Connect, Sensor Evaluation, Advanced Features, Data Analysis, AlgoBuilder, and Firmware Programming. The 'Advanced Features' section is expanded, showing options like Pedometer, FSM, MLC, Merge MLC + FSM, ISPU IDE, ISPU Model Converter, IR Algorithms Tuning, Configuration Converter, and HSDatalog Converter. The main workspace shows the 'FSM parameters' configuration for the ISM6H3256X sensor. It includes settings for State machine (#1), ODR (240 Hz), Converter (Float: 1.00, Float To Float16, Float16: 3C00, Float16 To Float), Long Counter (Max value: 0000), and FSM start address (FSM\_START\_H: 04, FSM\_START\_L: 00). Below these are tabs for ALL, CFG, and FSM. The FSM diagram shows states S0 (0x0A) and S1 (0x0B) with transitions and actions. A table lists the FSM states:

| Index | Address | Type | Value   |
|-------|---------|------|---------|
| S0    | 0x0A    | RNC  | NOP     |
| S1    | 0x0B    | CMD  | CONTROL |

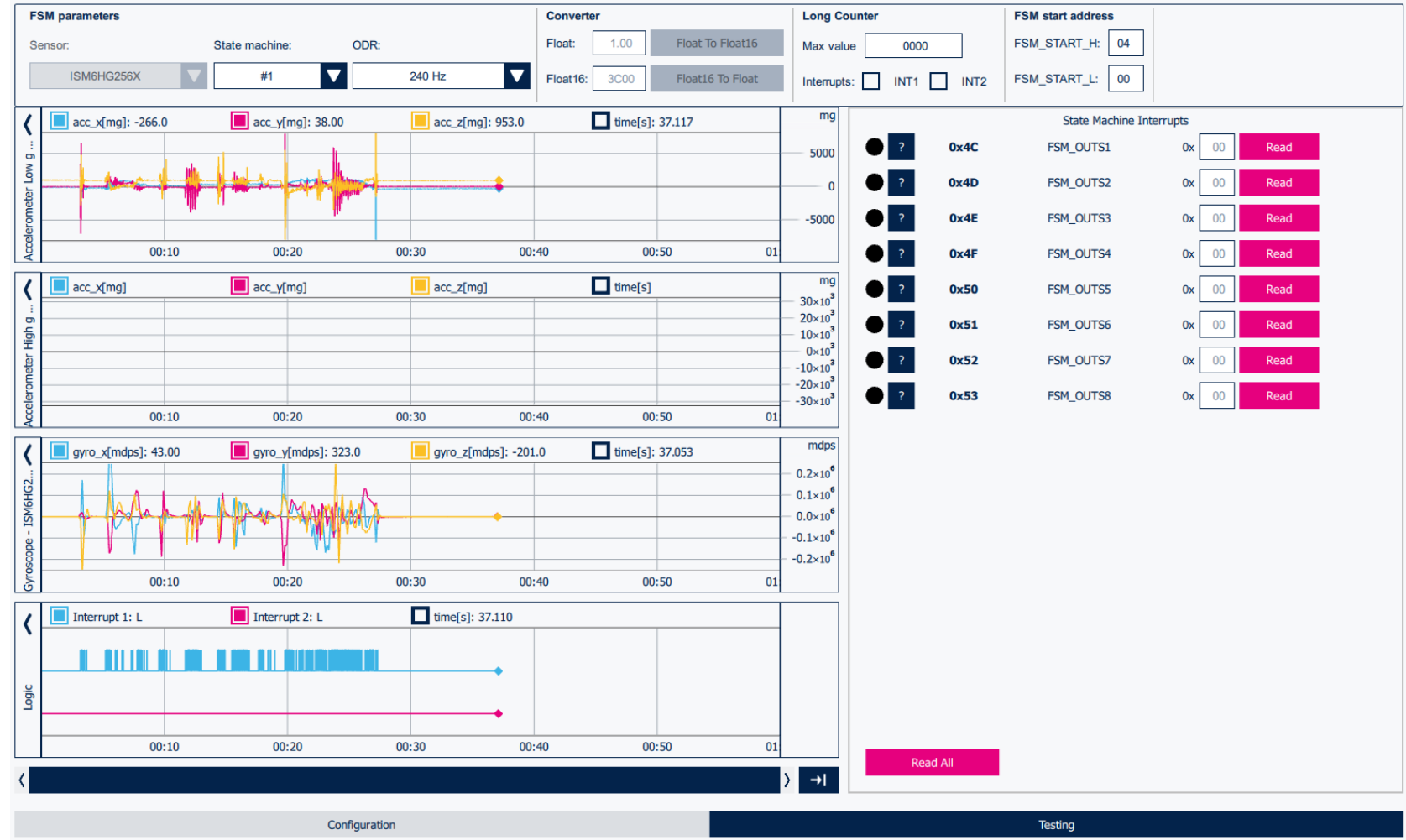
On the right side of the interface, there are sections for Status Data (Enabled, INT1, INT2), Fixed Data (Decimation), and Variable Data (Ox06 Thresh 1, Ox08 Mask A, Ox09 TempMask A). At the bottom, there are buttons for Configuration and Testing.

# HANDS ON

Once the data is streaming you can click on the Testing tab at the bottom of the UI.

You can move the Device to generate the interrupt.

# Testing



# Implementing Adaptive Self-Configuration (High-G)

# Adaptive self-configuration (ASC)

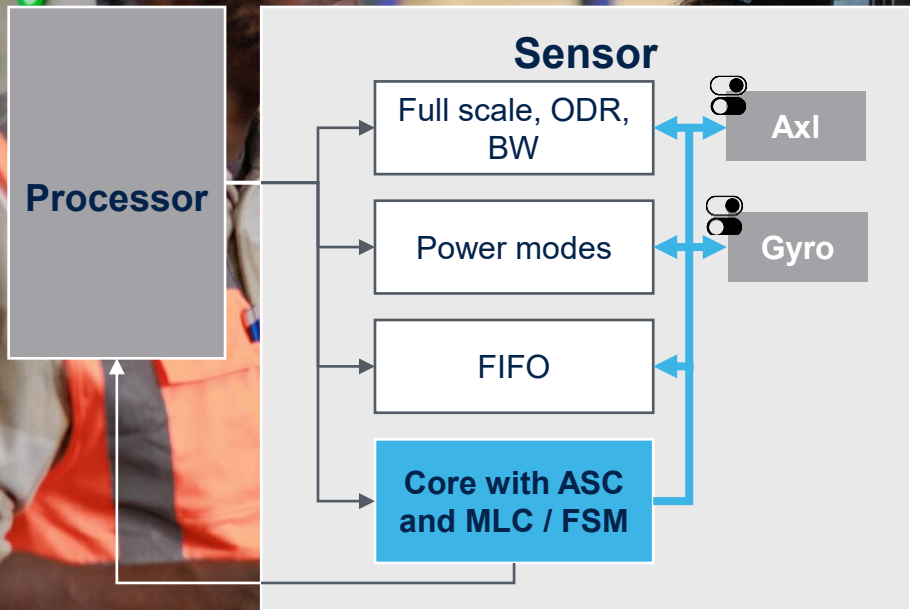
Always perfectly fits the context



The device automatically **reconfigures itself** based on the actual context, maximizing the **system efficiency**

**MLC Decision Tree** outputs can be routed to the **FSM Program** which controls the **ASC** feature

**ASC** allows independent configuration of gyroscope and accelerometer channels



# HANDS ON

We can navigate to the Sensor Evaluation Tab and go to the Quick Setup.

For this section we will be increasing our data rate to **480 Hz** and also enabling the high-g sensor

## Evaluating the High G Accel

Accelerometer low-g Full Scale:

8 g

Accelerometer low-g Operating Mode:

High-performance mode

Accelerometer low-g Output Data Rate:

480 Hz

Accelerometer high-g Full Scale:

32 g

Accelerometer high-g Output Data Rate:

480 Hz

# HANDS ON

You can now hit the play button in the top left and navigate to the Line Chart section of the Sensor Evaluation Tab to see your live data

If you are having trouble making out the High G data, you can navigate to the right-hand side of the graph and a menu will populate. Select Fit to have the graph scale to the data

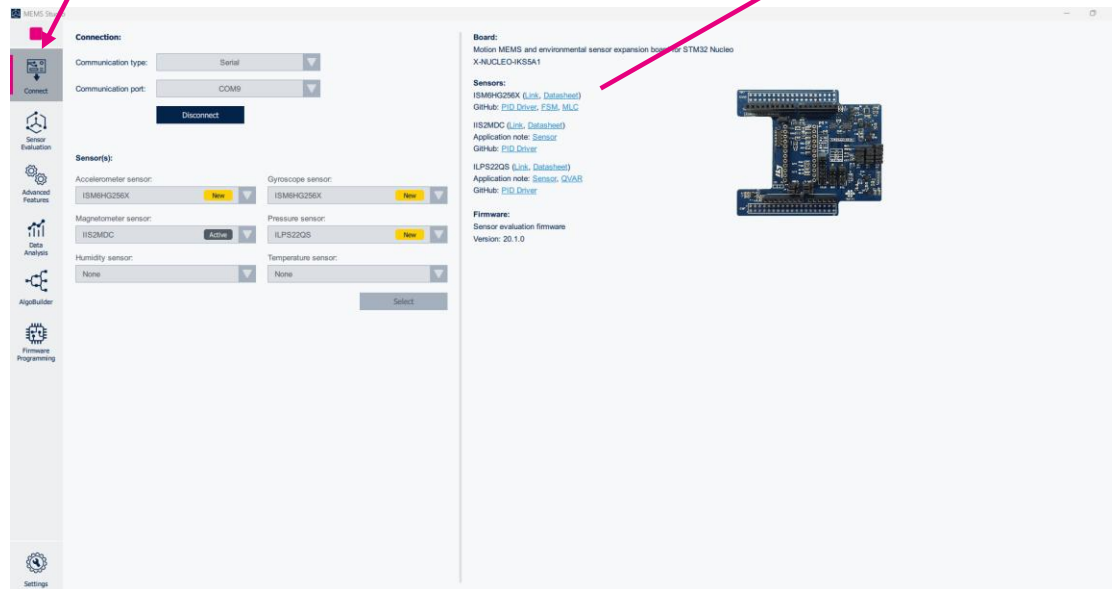
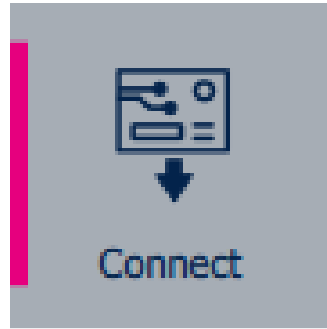


# HANDS ON

Before we go further you can spend some time downloading the documentation for the device.

We will be referencing it in the upcoming section, and it could be useful to build understanding.

To do this easily click on the connect tab and then the [Link](#) for the ISM6HG256X



## Documentation-Optional

### Board:

Motion MEMS and environmental sensor expansion board for STM32 Nucleo  
X-NUCLEO-IKS5A1

### Sensors:

ISM6HG256X ([Link](#), [Datasheet](#))

GitHub: [PID Driver](#), [FSM](#), [MLC](#)

HANDS  
ON

Select Documentation

# Documentation-Optional

ISM6HG256X

ACTIVE



 Save to myST

Intelligent IMU (inertial measurement unit) with simultaneous low-g and high-g acceleration detection



Download datasheet



Order Direct

Overview



Sample & Buy

Documentation

CAD Resources

Tools & Software

Quality & Reliability

## Product overview

Description

All features

You might also...

Recommended for you

### Description

The ISM6HG256X is a high-end, low-noise, low-power 6-axis IMU, featuring a 3-axis digital low-g accelerometer at 16 g, a 3-axis digital high-g accelerometer at 256 g, and a 3-axis digital gyroscope, which offers the best IMU sensor with a quad-channel architecture for processing acceleration and angular rate data on four separate channels (user interface, OIS, EIS, and high-g accelerometer data) with dedicated configuration, processing, and filtering along with a dedicated high-g sensor for high-g shock.

The ISM6HG256X enables edge computing, leveraging embedded advanced dedicated features such as a finite state machine (FSM) for configurable motion tracking and a machine learning core (MLC) for context awareness with exportable AI features for IoT applications.

The device supports the adaptive self-configuration (ASC) feature, which allows automatically reconfiguring the device in real time based on the detection of a specific motion pattern or based on the output of a specific decision tree configured in the MLC, without any intervention from the host processor.



# HANDS ON

## Documentation-Optional

We will be referencing AN6354 and AN6353

We will include page numbers and section information when relevant as well as screenshots in case you are unable to download the documents

### Application Notes (3)

|        |                                                                                       |     |             |                       |
|--------|---------------------------------------------------------------------------------------|-----|-------------|-----------------------|
| AN6354 | ISM6HG256X: finite state machine                                                      | 1.0 | 07 Oct 2025 | <a href="#">↓ PDF</a> |
| AN6355 | ISM6HG256X: machine learning core                                                     | 1.0 | 07 Oct 2025 | <a href="#">↓ PDF</a> |
| AN6353 | ISM6HG256X: intelligent IMU with simultaneous low-g and high-g acceleration detection | 1.0 | 02 Oct 2025 | <a href="#">↓ PDF</a> |

# HANDS ON

To navigate to the FSM tool go to the Advanced Features tab on the left side of the UI and then Click on the FSM option

# FSM Tool

The screenshot displays the MEMS Studio FSM Tool interface. On the left, a vertical sidebar contains icons for Connect, Sensor Evaluation, Advanced Features (highlighted), Data Analysis, AlgoBuilder, Firmware Programming, and Settings. The Advanced Features section is expanded, showing a list of options: Pedometer, FSM (selected), MLC, Merge MLC + FSM, ISPU IDE, ISPU Model Converter, IR Algorithms Tuning, Configuration Converter, and HSDatalog Converter.

The main workspace is titled 'FSM parameters' and includes several configuration sections:

- Sensor:** A dropdown menu showing 'ISM6HGX256X'.
- State machine:** A dropdown menu showing '#1'.
- ODR:** A dropdown menu showing '30 Hz'.
- Converter:** Fields for 'Float' (1.00) and 'Float16' (3C00), with buttons for 'Float To Float16' and 'Float16 To Float'.
- Long Counter:** A 'Max value' field set to '0000' and 'Interrupts' checkboxes for 'INT1' and 'INT2'.
- FSM start address:** Fields for 'FSM\_START\_H:' (04) and 'FSM\_START\_L:' (00).

Below these parameters is a large table with columns 'Index', 'Address', and 'Type'. The table is currently empty. To the right of the table, there are two sections: 'Status Data' with checkboxes for 'Enabled', 'INT1', and 'INT2', and 'Fixed Data' with a checkbox for 'Decimation'.

At the bottom of the interface, there is a 'Configuration' bar and a 'Testing' bar.

HANDS  
ON

# Set ODR

First step is to set the  
FSM Cores ODR. In this  
example we will be using  
**480Hz**

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 480 Hz

MEMS Studio

**Advanced Features**

- Pedometer
- **FSM**
- MLC
- Merge MLC + FSM
- ISPU IDE
- ISPU Model Converter
- IR Algorithms Tuning
- Configuration Converter
- HSDatalog Converter

Connect

Sensor Evaluation

Advanced Features

Data Analysis

AlgoBuilder

Firmware Programming

Settings

**FSM parameters**

Sensor: ISM6HG256X State machine: #1 ODR: 480 Hz

Converter

Float: 1.00 Float To Float16

Float16: 3000 Float16 To Float

Long Counter

Max value: 0000

Interrupts: ☐ INT1 ☐ INT2

FSM start address

FSM\_START\_H: 04

FSM\_START\_L: 00

Status Data

Enabled ☐

INT1 ☐

INT2 ☐

Fixed Data

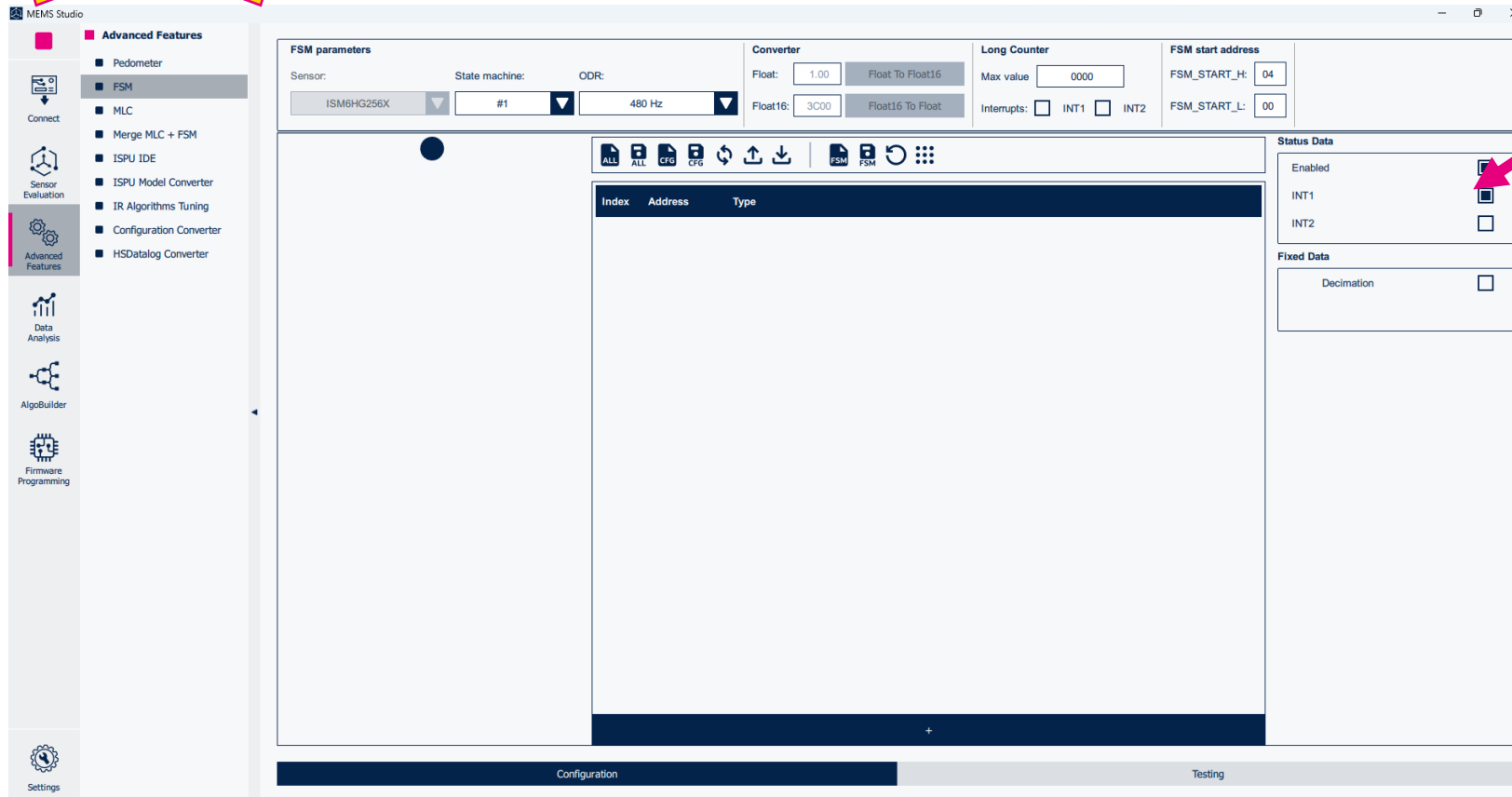
Decimation ☐

Index Address Type

Configuration Testing

HANDS  
ON

# Enable Interrupt and Status



**Status Data**

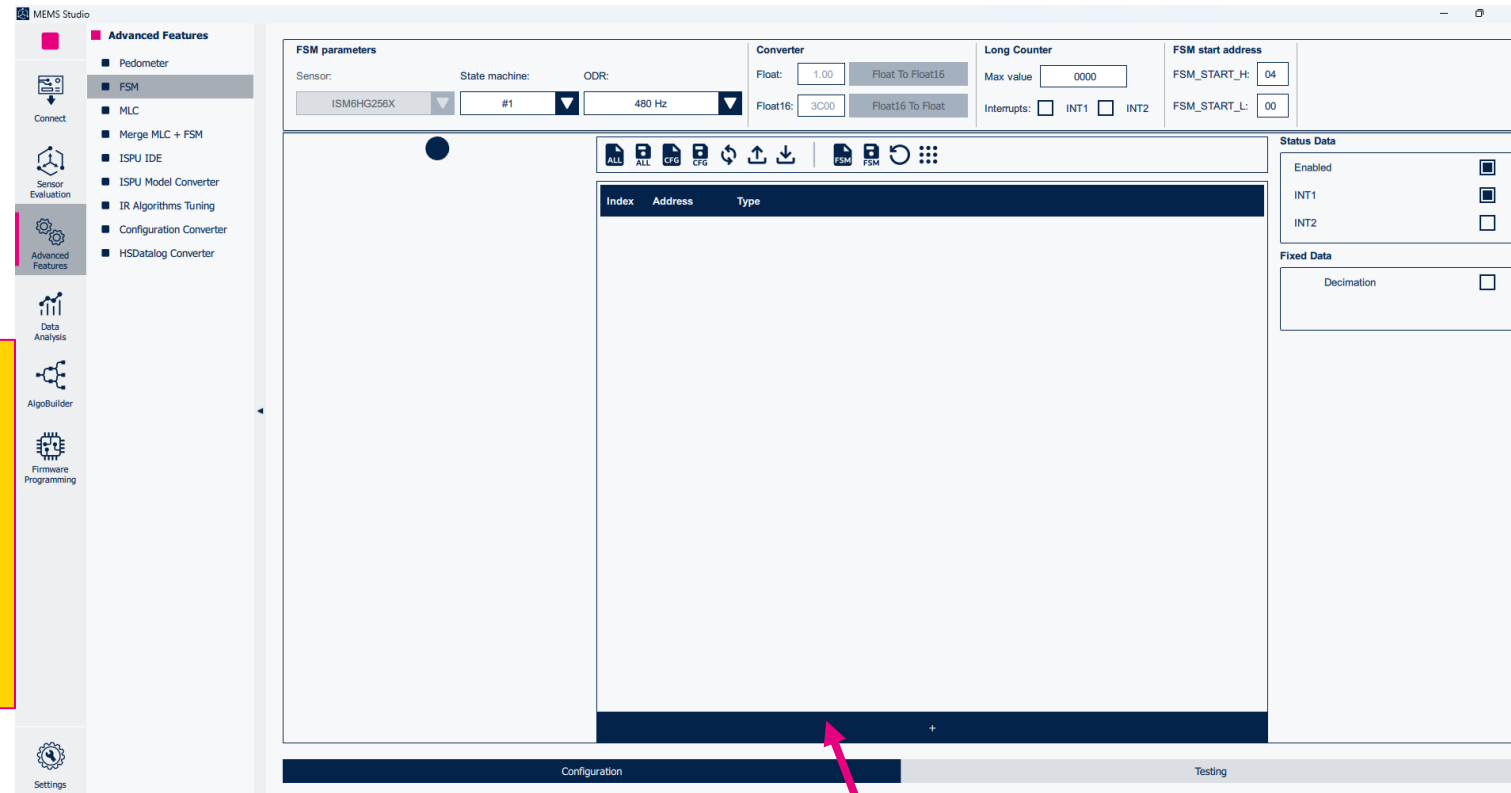
|         |                          |
|---------|--------------------------|
| Enabled | <input type="checkbox"/> |
| INT1    | <input type="checkbox"/> |
| INT2    | <input type="checkbox"/> |

Next, we will Enable the  
FSM Status and drive  
the interrupt to the INT1  
Interrupt pin

# HANDS ON

To start creating our FSM Program we need to add our First Instruction. To do so click the + button at the bottom of the UI

## Add New State



## HANDS ON

The SETR command is used to enable the adaptive self-configuration (**ASC**) feature, which allows the device to reconfigure itself without any intervention from the host processor.

To utilize this command, we need to know the register address we wish to change and the value we are changing it to

# The Set Register Command

|      |                 |       |                                 |  |   |   |   |
|------|-----------------|-------|---------------------------------|--|---|---|---|
| S0   | 0x06            | CMD ▼ | SETR ▼                          |  | 📄 | + | - |
| 0x07 | REG_ADDR Value: | 0x    | <input type="text" value="00"/> |  |   |   |   |
| 0x08 | REG_VAL Value:  | 0x    | <input type="text" value="00"/> |  |   |   |   |

# HANDS ON

To toggle and customize the High G accelerometer we need to write to its control register. This register is labeled **CTRL1\_XL\_HG** and has the Register address of **13h** when referenced from the FSM

(This table is found on page 40 section 10.2.7 of the AN6354)

## The High G Accelerometer Register

Table 10. ASC FSM main page registers

| Main page register | Register address (controller) | Register address (FSM) |
|--------------------|-------------------------------|------------------------|
| FIFO_CTRL1         | 07h                           | 07h                    |
| FIFO_CTRL2         | 08h                           | 08h                    |
| FIFO_CTRL3         | 09h                           | 09h                    |
| FIFO_CTRL4         | 0Ah                           | 0Ah <sup>(1)</sup>     |
| CTRL1              | 10h                           | 10h                    |
| CTRL2              | 11h                           | 11h                    |
| CTRL1_XL_HG        | 4Eh                           | 13h                    |
| CTRL2_XL_HG        | 4Dh                           | 14h                    |
| CTRL6              | 15h                           | 15h                    |
| CTRL7              | 16h                           | 16h                    |
| CTRL8              | 17h                           | 17h                    |
| CTRL9              | 18h                           | 18h                    |
| CTRL10             | 19h                           | 19h                    |

# HANDS ON

## CTRL1\_XL\_HG

The CTRL1\_XL\_HG register contains bits that control the Full Scale[2:0], Output Data Rate[5:3] and Register Out toggle[7] for the High G accelerometer

More information can be found on page 20 section 3.2 of the AN6353

CTRL1\_XL\_HG

4Eh

XL\_HG\_REGOUT\_EN

HG\_USR\_OFF\_ON\_OUT

ODR\_XL\_HG\_2

ODR\_XL\_HG\_1

ODR\_XL\_HG\_0

FS\_XL\_HG\_2

FS\_XL\_HG\_1

FS\_XL\_HG\_0

| Register name | Address | Bit7            | Bit6              |
|---------------|---------|-----------------|-------------------|
| CTRL1_XL_HG   | 4Eh     | XL_HG_REGOUT_EN | HG_USR_OFF_ON_OUT |

| Bit5        | Bit4        | Bit3        |
|-------------|-------------|-------------|
| ODR_XL_HG_2 | ODR_XL_HG_1 | ODR_XL_HG_0 |

| Bit2       | Bit1       | Bit0       |
|------------|------------|------------|
| FS_XL_HG_2 | FS_XL_HG_1 | FS_XL_HG_0 |



## HANDS ON

Since we want the sensor to be running at 480Hz and 32g

For **480Hz** we want a value of **011**

For **32g** we want a value of **000**

## Determining the Hex Value

| ODR_XL_HG [2:0] | ODR selection [Hz]   |
|-----------------|----------------------|
| 000             | Power-down (default) |
| 001             | Reserved             |
| 010             | Reserved             |
| 011             | 480                  |
| 100             | 960                  |
| 101             | 1920                 |
| 110             | 3840                 |
| 111             | 7680                 |

| FS_XL_HG [2:0] | Full Scale selection [g] |
|----------------|--------------------------|
| 000            | 32                       |
| 001            | 64                       |
| 010            | 128                      |
| 011            | 256                      |

HANDS  
ON

# CTRL1\_XL\_HG

| Bit7            | Bit6              | Bit5        | Bit4        | Bit3        | Bit2       | Bit1       | Bit0       |
|-----------------|-------------------|-------------|-------------|-------------|------------|------------|------------|
| XL_HG_REGOUT_EN | HG_USR_OFF_ON_OUT | ODR_XL_HG_2 | ODR_XL_HG_1 | ODR_XL_HG_0 | FS_XL_HG_2 | FS_XL_HG_1 | FS_XL_HG_0 |
| 1               | 0                 | 0           | 1           | 1           | 0          | 0          | 0          |

1001 = 9

98h

1000 = 8

Now putting everything together we can calculate the value to write to the register to enable the High G accelerometer output at 480Hz at 32g. Converting into Binary and the Hex we get the Value of **98h**

## HANDS ON

We can now update our SETR Command with the values we just calculated.

For REG\_ADDR Value we will be using the FSM Register Address for the CTRL1\_XL\_HG which is **13h**

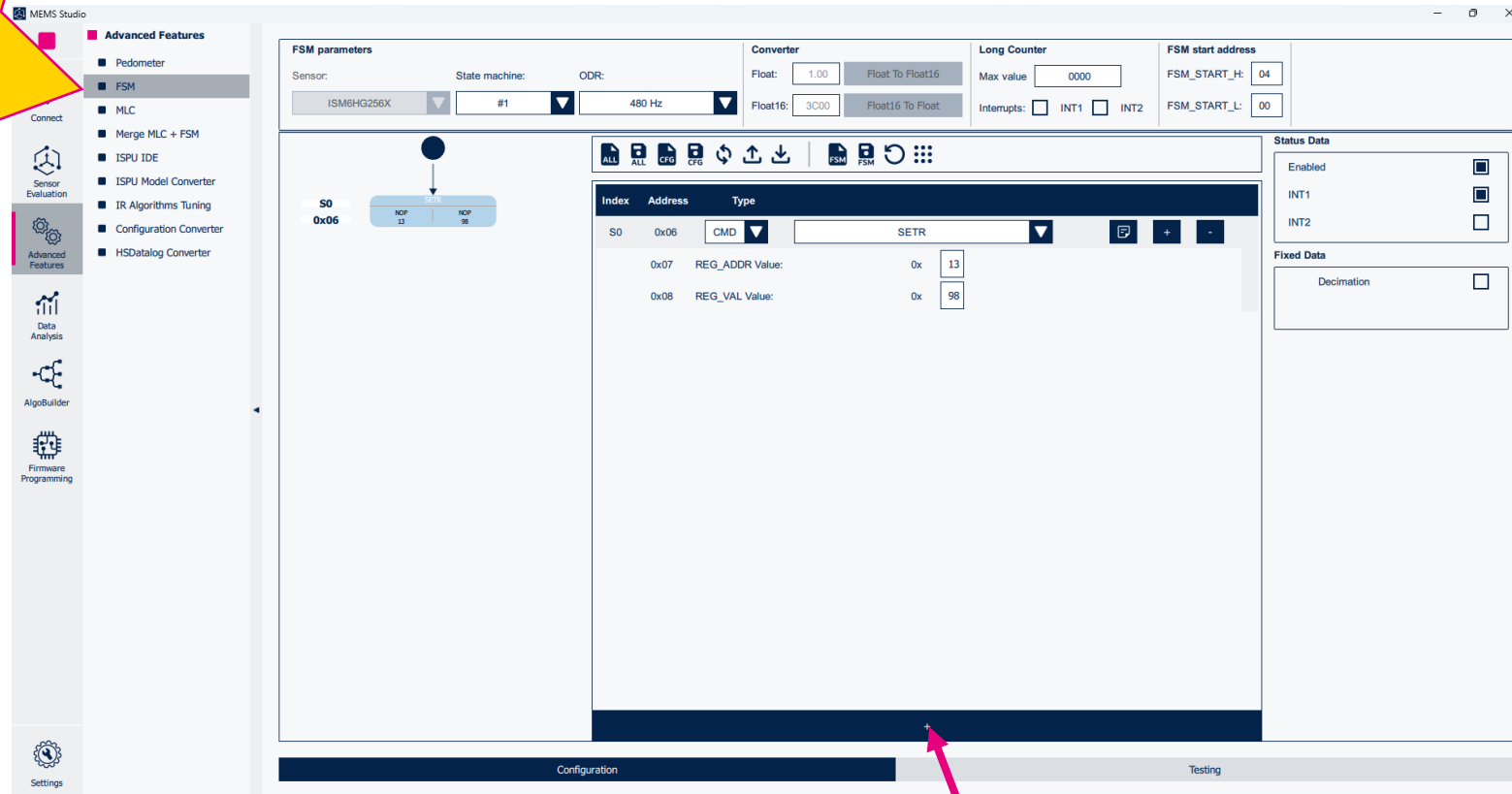
For the REG\_VAL value we will be using the **98h** value we calculated to correspond to 480Hz at 32g

# The Set Register Command

|      |                 |       |                                 |  |   |   |   |
|------|-----------------|-------|---------------------------------|--|---|---|---|
| S0   | 0x06            | CMD ▼ | SETR ▼                          |  | 📄 | + | - |
| 0x07 | REG_ADDR Value: | 0x    | <input type="text" value="13"/> |  |   |   |   |
| 0x08 | REG_VAL Value:  | 0x    | <input type="text" value="98"/> |  |   |   |   |

HANDS  
ON

# Add New State



The image shows the MEMS Studio interface for configuring a Finite State Machine (FSM). The left sidebar contains navigation options: Connect, Sensor Evaluation, Advanced Features (selected), Data Analysis, AlgoBuilder, and Firmware Programming. The Advanced Features section lists: Pedometer, FSM (selected), MLC, Merge MLC + FSM, ISPU IDE, ISPU Model Converter, IR Algorithms Tuning, Configuration Converter, and HSDatalog Converter. The main workspace is divided into several panels. The top panel shows FSM parameters: Sensor (ISM6HG256X), State machine (#1), ODR (480 Hz), Converter (Float: 1.00, Float16: 3C00), Long Counter (Max value: 0000, Interrupts: INT1, INT2), and FSM start address (FSM\_START\_H: 04, FSM\_START\_L: 00). Below this is a state transition diagram showing a state S0 at address 0x06 with two NOP instructions. The central panel is a table of instructions:

| Index | Address         | Type  |
|-------|-----------------|-------|
| S0    | 0x06            | CMD   |
|       |                 | SETR  |
| 0x07  | REG_ADDR Value: | 0x 13 |
| 0x08  | REG_VAL Value:  | 0x 98 |

The right panel shows Status Data (Enabled, INT1, INT2) and Fixed Data (Decimation). At the bottom, there are tabs for Configuration and Testing. A red arrow points from the bottom of the instruction table to a magnified view below.

Let us add another  
Instruction



HANDS  
ON

# Add Reset Next Condition

|    |      |       |         |       |                                                                                     |   |   |
|----|------|-------|---------|-------|-------------------------------------------------------------------------------------|---|---|
| S1 | 0x11 | RNC ▼ | GNTH1 ▼ | TI1 ▼ |  | + | - |
|----|------|-------|---------|-------|-------------------------------------------------------------------------------------|---|---|

This time instead of determining if the device is in motion we want to check if the device is still

To accomplish this, we will use the Greater than Threshold 1 or **GNTH1** as our reset condition to cause the program to reset if in motion

We don't want it instantly saying the device is still if the threshold falls below a value so we will utilize a timer. In this case Timer 1 or **TI1**

# HANDS ON

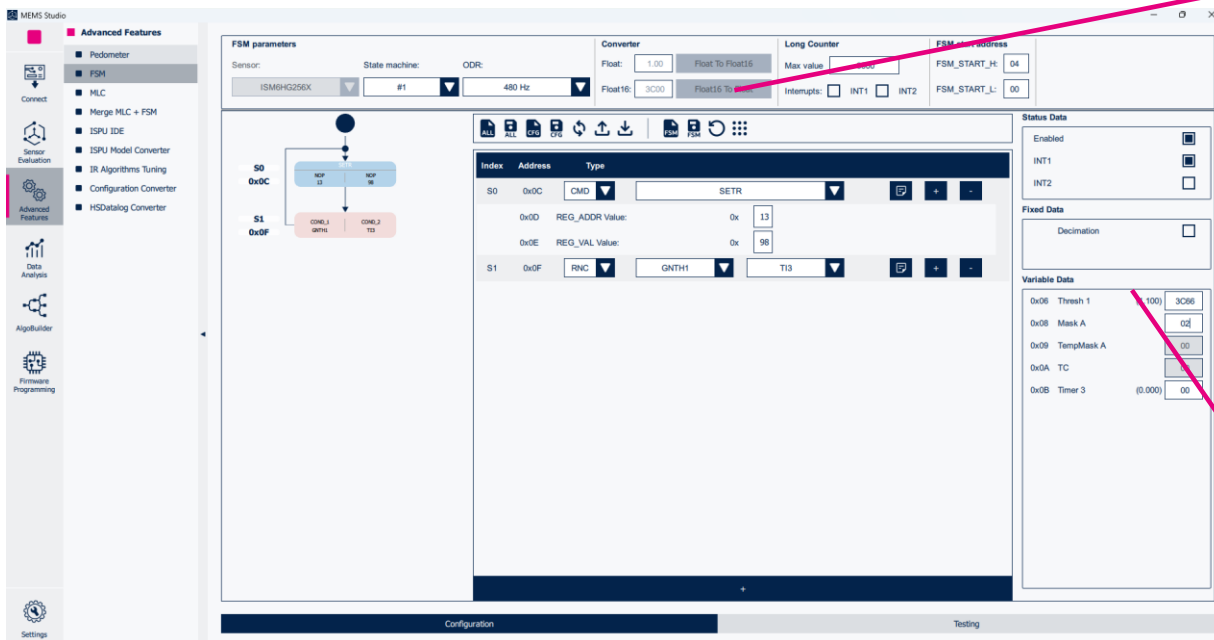
## Update Threshold Using Converter

Now we need to set the value for the variable Thresh 1.

We want to use 1.1g as the threshold.

The Variable however is expecting a Float16 Hex value.

You can generate this using the converter at the top of the UI and coping the result into the variable Section. You will see the UI update with the converted value.



**Converter**

Float:

Float16:

**Variable Data**

|      |          |         |      |
|------|----------|---------|------|
| 0x06 | Thresh 1 | (1.100) | 3C66 |
|------|----------|---------|------|

# HANDS ON

## Set Mask A to V+

Set Mask A

$$X+ \mid X- \mid Y+ \mid Y- \mid Z+ \mid Z- \mid V+ \mid V- = 0000 \ 0010 = 02$$

Binary Hex

Next, we want to set the Mask Value.

We can utilize If you hover over the Mask A variable it will show the mask selection for the current part.

### Variable Data

|      |          |         |      |
|------|----------|---------|------|
| 0x06 | Thresh 1 | (1.100) | 3C66 |
| 0x08 | Mask A   |         | 02   |

We want to use the Norm of all the axis, so this works regardless of orientation and motion.

Therefore, we want to use **V+** or **02** in Hex

# HANDS ON

## FSM Timers

We want to wait about 2 seconds to make the device is truly still.

2 seconds at 480 Hz is 960 samples. Converted to Hex that is **03C0** samples

In an actual application you would probably want a higher value but for the purpose of a snappy demo 2 seconds is a nice balance

### Variable Data

|      |            |         |      |
|------|------------|---------|------|
| 0x06 | Thresh 1   | (1.100) | 3C66 |
| 0x08 | Mask A     |         | 02   |
| 0x09 | TempMask A |         | 00   |
| 0x0A | TC         |         | 0000 |
| 0x0C | Timer 1    | (2.000) | 03C0 |



HANDS  
ON

# Add New State

The MEMS Studio interface displays the FSM configuration for the ISM6HG256X sensor. The left sidebar shows the 'Advanced Features' menu with options like Pedometer, FSM, MLC, and others. The main window shows the FSM parameters, including the sensor, state machine, ODR, and converter settings. The FSM diagram shows a sequence of states: S0 (NOP), S1 (NOP), and S2 (NOP). The table below shows the instructions for each state.

| Index | Address         | Type | Instruction |
|-------|-----------------|------|-------------|
| S0    | 0x0E            | CMD  | SETR        |
| 0x0F  | REG_ADDR Value: | 0x   | 13          |
| 0x10  | REG_VAL Value:  | 0x   | 98          |
| S1    | 0x11            | RNC  | GNTN1       |
|       |                 |      | TI1         |

Let us add another  
Instruction



## HANDS ON

This time we will be using the ASC set register command to turn the High-G sensor off.

For REG\_ADDR Value we will be using the FSM Register Address for the CTRL1\_XL\_HG which is **13h**

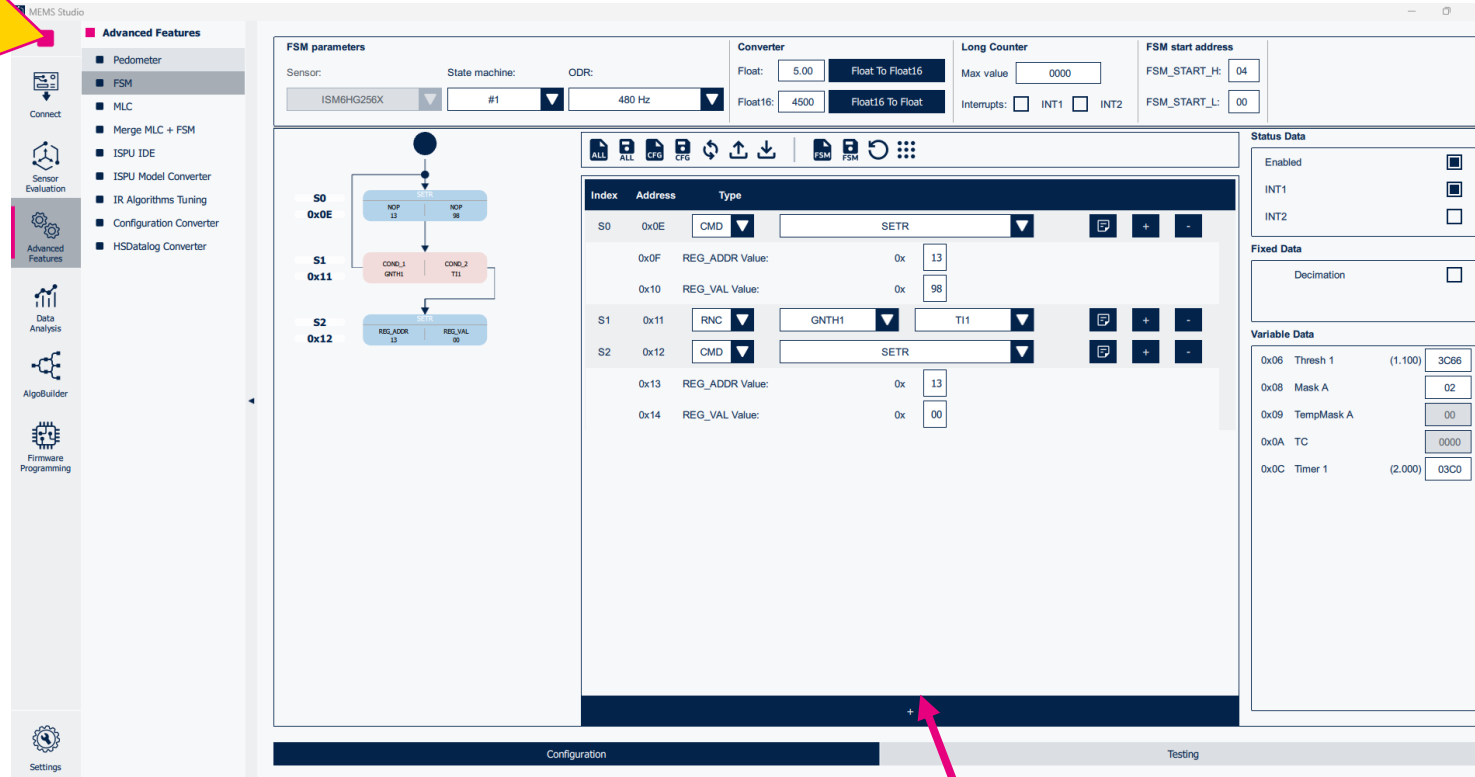
For the REG\_VAL value we will be using the **00h** value to put the sensor in Power Down mode to save power

# The Set Register Command

|    |      |                 |        |    |   |   |   |
|----|------|-----------------|--------|----|---|---|---|
| S2 | 0x12 | CMD ▼           | SETR ▼ |    | 📄 | + | - |
|    | 0x13 | REG_ADDR Value: | 0x     | 13 |   |   |   |
|    | 0x14 | REG_VAL Value:  | 0x     | 00 |   |   |   |

HANDS  
ON

# Add New State



The image shows the MEMS Studio interface for configuring a Finite State Machine (FSM). The left sidebar contains various tool icons, with 'Advanced Features' selected. The main window is divided into several sections:

- FSM parameters:** Includes fields for Sensor (ISM6HG256X), State machine (#1), ODR (480 Hz), Converter (Float: 5.00, Float16: 4500), Long Counter (Max value: 0000), and FSM start address (FSM\_START\_H: 04, FSM\_START\_L: 00).
- FSM Diagram:** A state transition diagram showing three states: S0 (0x0E), S1 (0x11), and S2 (0x12). Transitions are labeled with instructions like NOP, CORN, and REG\_ADDR.
- Instruction Table:** A table with columns for Index, Address, and Type. It lists instructions for S0, S1, and S2, including SETR, RNC, and CORN.
- Status Data:** Includes checkboxes for Enabled, INT1, and INT2.
- Fixed Data:** Includes a checkbox for Decimation.
- Variable Data:** Includes fields for Thresh 1, Mask A, TempMask A, TC, and Timer 1.

A red arrow points from the '+' button at the bottom of the instruction table to a larger '+' button in a separate window below.

Let us add another  
Instruction

HANDS  
ON

# Add Reset Next Condition for Motion

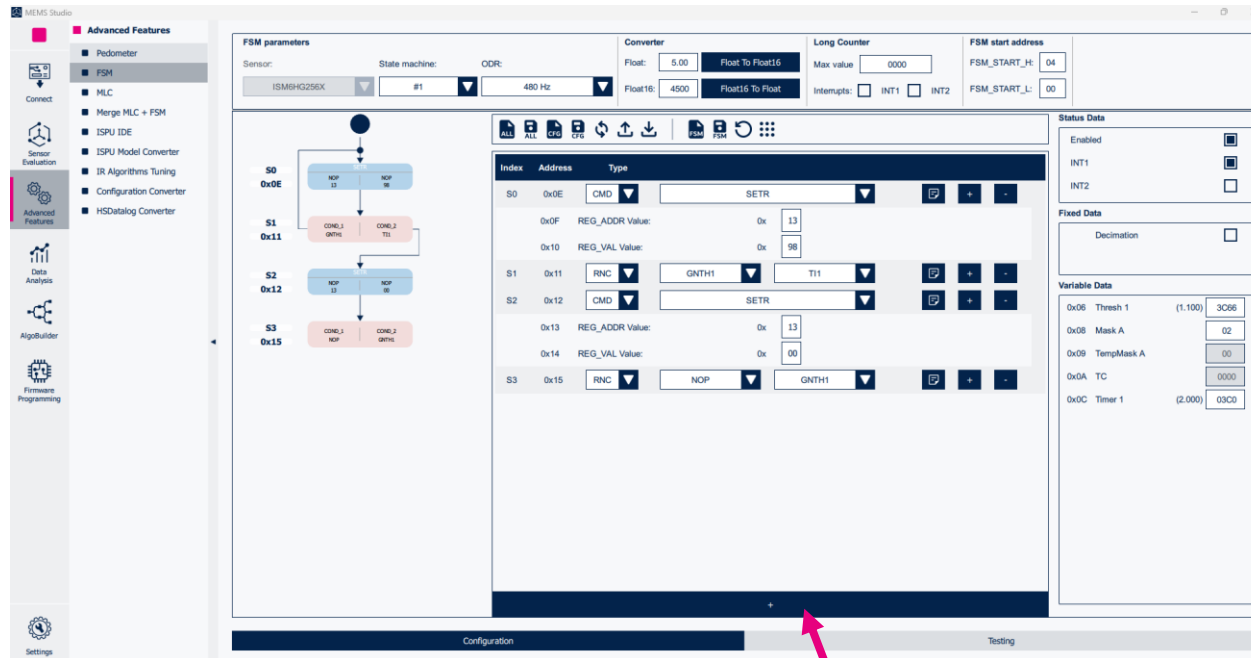


We want to move out of this state and reenale the High G accelerometer the second we have motion

This will loop forever until the Next condition is true. In this case we will use the Greater than Threshold 1 or **GNTH1**

HANDS  
ON

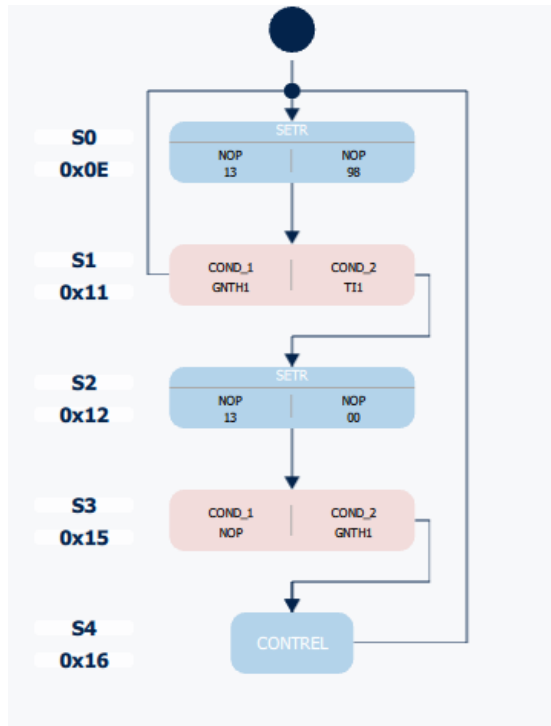
# Add New State



Let us add another  
Instruction



# HANDS ON



## Set to Command CONTREL

| Index | Address | Type            |         |       |  |  |  |  |
|-------|---------|-----------------|---------|-------|--|--|--|--|
| S0    | 0x0E    | CMD             | SETR    |       |  |  |  |  |
|       | 0x0F    | REG_ADDR Value: | 0x      | 13    |  |  |  |  |
|       | 0x10    | REG_VAL Value:  | 0x      | 98    |  |  |  |  |
| S1    | 0x11    | RNC             | GNTH1   | TI1   |  |  |  |  |
| S2    | 0x12    | CMD             | SETR    |       |  |  |  |  |
|       | 0x13    | REG_ADDR Value: | 0x      | 13    |  |  |  |  |
|       | 0x14    | REG_VAL Value:  | 0x      | 00    |  |  |  |  |
| S3    | 0x15    | RNC             | NOP     | GNTH1 |  |  |  |  |
| S4    | 0x16    | CMD             | CONTREL |       |  |  |  |  |

We will use the command CONTREL. This resets the program from the Reset Pointer and generates an Interrupt

HANDS  
ON

# Save Configuration

We now can save the  
configuration



The MEMS Studio software interface is shown. The 'FSM parameters' window is open, displaying a state machine diagram on the left and a table of FSM parameters on the right. The state machine diagram shows states S0 through S4 with transitions and actions. The table lists the index, address, and type of each FSM entry.

| Index | Address | Type |
|-------|---------|------|
| S0    | 0x0E    | CMD  |
| S1    | 0x0F    | RNC  |
| S2    | 0x10    | CMD  |
| S3    | 0x13    | RNC  |
| S4    | 0x15    | CMD  |

The interface also includes sections for 'Advanced Features', 'Sensor Evaluation', 'Data Analysis', 'AlgoBuilder', 'Firmware Programming', and 'Settings'. The 'Status Data' section on the right shows various status indicators and data points.

HANDS  
ON

# Write Configuration to Sensor



And write the  
configuration to the  
device

The screenshot shows the MEMS Studio software interface. On the left is a sidebar with icons for Connect, Sensor Evaluation, Advanced Features, Data Analysis, Algorithm Builder, and Settings. The 'Advanced Features' section is expanded, showing options like Pedometer, FSM, MLC, Merge MLC + FSM, ISPU IDE, ISPU Model Converter, IR Algorithms Tuning, Configuration Converter, and HSDatalog Converter. The main window displays the 'FSM parameters' for the ISM6HG256X sensor. It includes fields for State machine (#1), ODR (480 Hz), Converter (Float: 1.00, Float To Float16, Float16: 3C00, Float16 To Float), Long Counter (Max value: 0000, Interrupts: INT1, INT2), and FSM start address (FSM\_START\_H: 04, FSM\_START\_L: 00). Below these is a state machine diagram with states S0 to S4. To the right of the diagram is a table for configuration parameters:

| Index | Address | Type            | Value   | Buttons               |
|-------|---------|-----------------|---------|-----------------------|
| S0    | 0x0E    | CMD             | SETR    | [Copy] [Add] [Remove] |
|       | 0x0F    | REG_ADDR Value: | 0x 13   |                       |
|       | 0x10    | REG_VAL Value:  | 0x 98   |                       |
| S1    | 0x11    | RNC             | GNT1    | [Copy] [Add] [Remove] |
| S2    | 0x12    | CMD             | SETR    | [Copy] [Add] [Remove] |
|       | 0x13    | REG_ADDR Value: | 0x 00   |                       |
|       | 0x14    | REG_VAL Value:  | 0x 00   |                       |
| S3    | 0x15    | RNC             | NOP     | [Copy] [Add] [Remove] |
| S4    | 0x16    | CMD             | CONTROL | [Copy] [Add] [Remove] |

At the bottom of the main window are tabs for 'Configuration' and 'Testing'. The 'Write to Sensor' icon from the toolbar is highlighted with a red box and an arrow pointing to the 'Write' button in the configuration table.



# HANDS ON

# Enable ASC

Navigate to the Sensor Evaluation tab and in the drop-down menu for Registers Map click ISM6HG256X

MEMS Studio

**Sensor Evaluation**

- Quick Setup
- Registers Map
- ISM6HG256X
- IIS2MDC
- ILPS22QS
- Save to File
- Data Table
- Data Monitor
- SM Monitor
- LC Monitor
- Charts
- Scatter Plot
- Interrupt
- OT
- Load/Save Configuration
- Sample Configurations

Connect

Sensor Evaluation

Programming

Settings

| Basic Registers |                  |       |      |       |         | Embedded Functions Registers |    |    |    |    |    | Advanced Features Registers Page 0 |    |         |                     |       |      | Advanced Features Registers Page 1 |         |    |    |    |    | Advanced Features Registers Page 2 |    |    |    |  |  | Sensor Hub Registers |  |  |  |  |  |
|-----------------|------------------|-------|------|-------|---------|------------------------------|----|----|----|----|----|------------------------------------|----|---------|---------------------|-------|------|------------------------------------|---------|----|----|----|----|------------------------------------|----|----|----|--|--|----------------------|--|--|--|--|--|
| Address         | Register Name    | Value |      |       |         | B7                           | B6 | B5 | B4 | B3 | B2 | B1                                 | B0 | Address | Register Name       | Value |      |                                    |         | B7 | B6 | B5 | B4 | B3                                 | B2 | B1 | B0 |  |  |                      |  |  |  |  |  |
| 0x01            | FUNC_CFG_ACCESS  | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x43    | TIMESTAMP3          | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x02            | PIN_CTRL         | 0x 23 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x44    | UI_STATUS_REG_OIS   | 0x 04 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x03            | IF_CFG           | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x45    | WAKE_UP_SRC         | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x06            | ODR_TRIG_CFG     | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x46    | TAP_SRC             | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x07            | FIFO_CTRL1       | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x47    | D6D_SRC             | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x08            | FIFO_CTRL2       | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x48    | STATUS_C...MAINPAGE | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x09            | FIFO_CTRL3       | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x49    | EMB_FUN...AINPAGE   | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0A            | FIFO_CTRL4       | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4A    | FSM_STAT...MAINPAGE | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0B            | COUNTER_BDR_REG1 | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4B    | MLC_STAT...MAINPAGE | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0C            | COUNTER_BDR_REG2 | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4C    | HG_WAKE_UP_SRC      | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0D            | INT1_CTRL        | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4D    | CTRL2_XL_HG         | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0E            | INT2_CTRL        | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4E    | CTRL1_XL_HG         | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x0F            | WHO_AM_I         | 0x 73 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x4F    | INTERNAL_FREQ_FINE  | 0x F0 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x10            | CTRL1            | 0x 08 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x50    | FUNCTIONS_ENABLE    | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x11            | CTRL2            | 0x 06 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x52    | HG_FUNC...S_ENABLE  | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x12            | CTRL3            | 0x 44 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x53    | HG_WAKE_UP_THS      | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x13            | CTRL4            | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x54    | INACTIVITY_DUR      | 0x 04 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x14            | CTRL5            | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x55    | INACTIVITY_THS      | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x15            | CTRL6            | 0x 0C | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x56    | TAP_CFG0            | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |
| 0x16            | CTRL7            | 0x 00 | Read | Write | Default |                              |    |    |    |    |    |                                    |    | 0x57    | TAP_CFG1            | 0x 00 | Read | Write                              | Default |    |    |    |    |                                    |    |    |    |  |  |                      |  |  |  |  |  |

Read all Write all Default all Custom Register Action

Registers with ODR value might be modified by firmware.

# Enable ASC

HANDS  
ON

Click the ? Button for **FUNC\_CFG\_ACCESS** to pull up the register information

| Address       | Register Name       | Value                                                                                                                                                                                                             | B7                       | B6                       | B5                       | B4                       | B3                                  | B2                       | B1                       | B0                       |
|---------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
| <b>?</b> 0x01 | FUNC_CFG_ACCESS     | 0x <b>08</b> <span>Read</span> <span>Write</span> <span>Default</span>                                                                                                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| [7]           | EMB_FUNC_CFG_ACCESS | Enable access to the embedded functions configuration registers. Default value: 0                                                                                                                                 |                          |                          |                          |                          |                                     |                          |                          |                          |
| [6]           | SHUB_REG_ACCESS     | Enable access to the sensor hub (I2C controller) registers. Default value: 0                                                                                                                                      |                          |                          |                          |                          |                                     |                          |                          |                          |
| [3]           | FSM_WR_CTRL_EN      | Enables the control of the CTRL registers to FSM (FSM can change some configurations of the device autonomously). Default value: 0 (0: disabled; 1: enabled)                                                      |                          |                          |                          |                          |                                     |                          |                          |                          |
| [2]           | SW_POR              | Global reset of the device. Default value: 0                                                                                                                                                                      |                          |                          |                          |                          |                                     |                          |                          |                          |
| [1]           | AUX_SPI_RESET       | Resets the control registers of SPI from the primary interface. This bit must be set to 1 and then back to 0 (this bit is not automatically cleared). Default value: 0                                            |                          |                          |                          |                          |                                     |                          |                          |                          |
| [0]           | OIS_CTRL_FROM_UI    | Enables the full control of OIS configurations from the primary interface. Default value: 0 (0: OIS chain full control from primary interface disabled; 1: OIS chain full control from primary interface enabled) |                          |                          |                          |                          |                                     |                          |                          |                          |

Since we want to give the FSM the ability to write to the control registers, we need to activate **FSM\_WR\_CTRL\_EN**

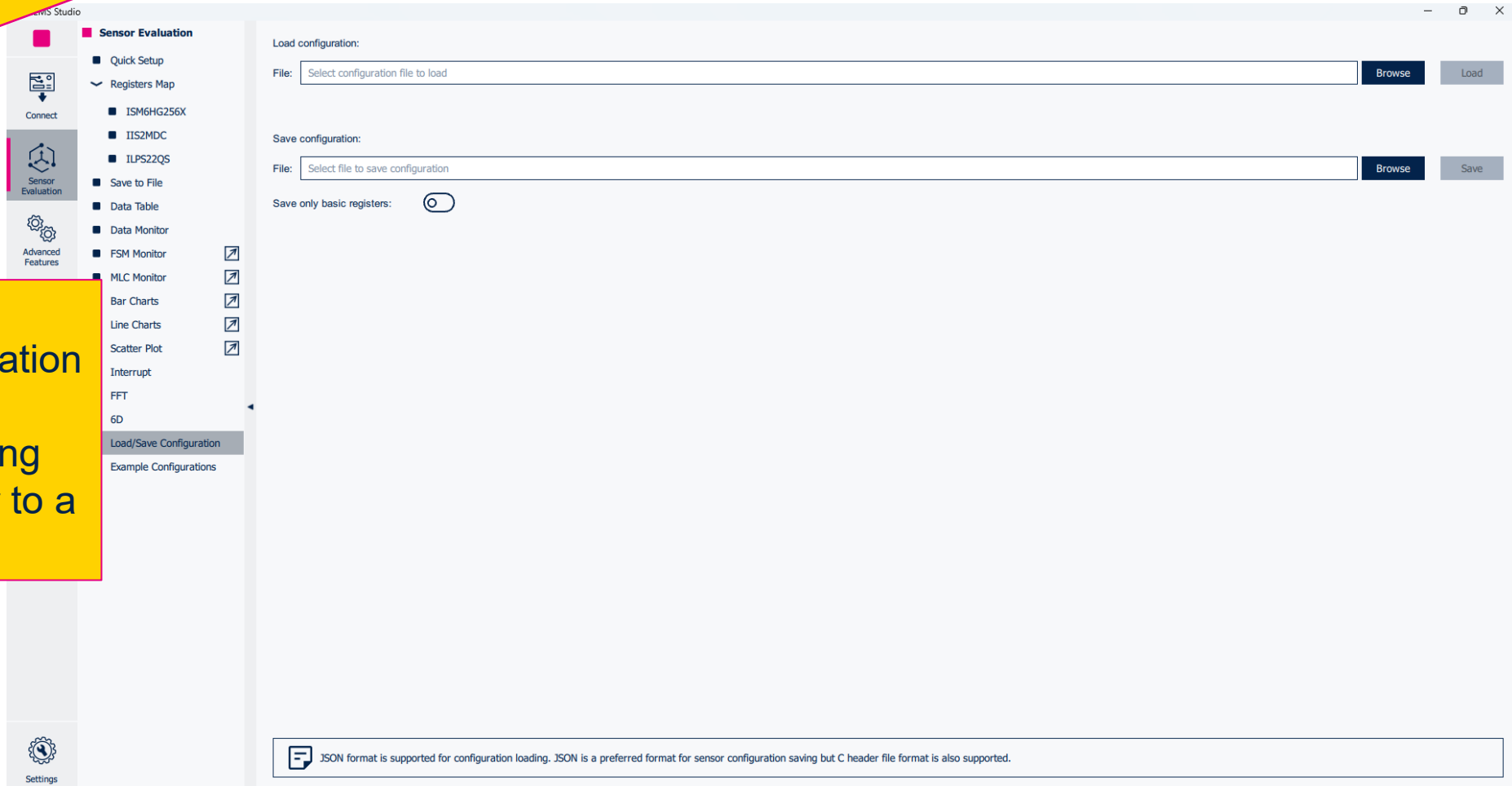
To do this click on the **B3** button in the UI to activate Bit 3

Once the UI correctly populates with the value **08** click **Write** to update the register

# HANDS ON

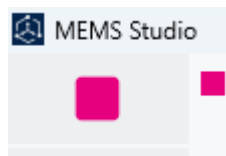
## Save Configuration

Navigate to the Load/Save Configuration button to save this configuration including the updated register to a .json file

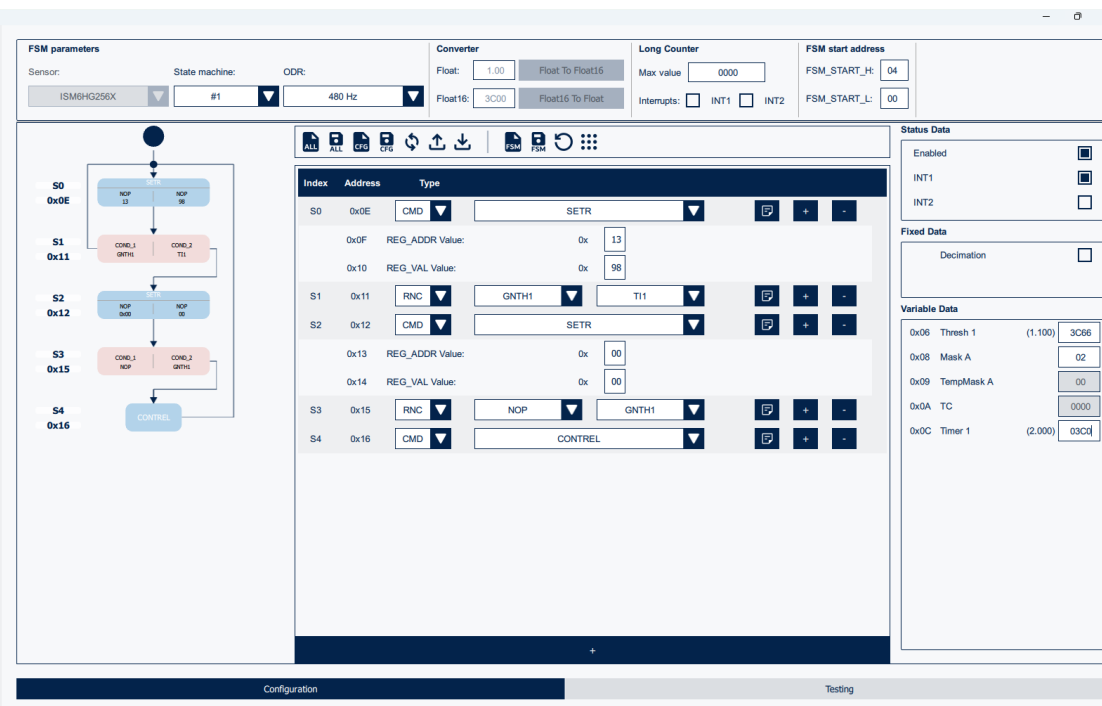


HANDS  
ON

# Start Data Streaming



Hit the play button in the top right of the UI to begin streaming data

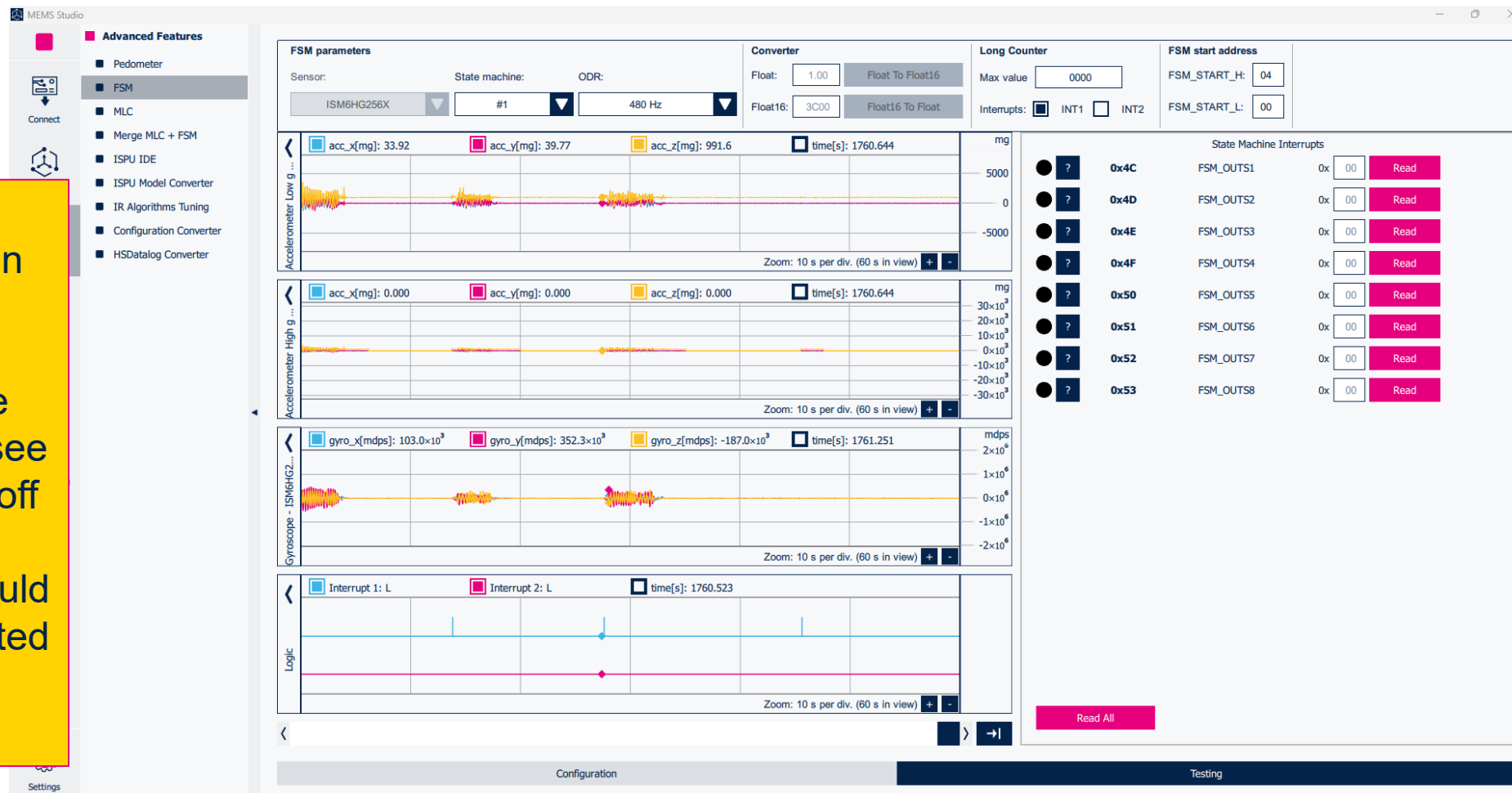


# HANDS ON

# Validate

We can now navigate back to the Testing tab in the FSM tool.

When you let the device rest for awhile you will see the High G sensor turn off and when you interact with the device you should see an interrupt generated and the High G sensor toggle on



# Questions or support needed?

## **ST Online Support Center**

Online support requests can be submitted <https://my.st.com/ols>

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